








# Modelling Financial Markets

## Chapter 13: Machine Learning in Finance



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# What Can Machine Learning Predict in Finance?



Campbell R. Harvey.
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*"Hundreds of papers and factors attempt to explain the cross-section of expected returns. Given this extensive data mining, it does not make sense to use the usual criteria for establishing significance."*

— *Harvey, Liu & Zhu (2016)*

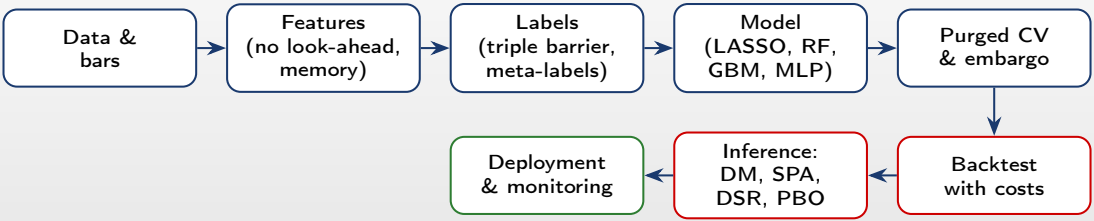
- After hundreds of factors tried on the same data, a new factor needs $t > 3$, not $t > 2$
- 5-day direction of the S&P 500, $n = 6{,}269$: no model beats the "always up" benchmark
  - the benchmark has 58.1% accuracy; the logit has 57.7% and predicts "up" on 98.3% of days
- Walk-forward backtest 2010–2026: the Sharpe ratio of the GBM strategy is 0.33
  - the Sharpe ratio of buy-and-hold is 0.66
- On a target with nothing to predict, shuffled K-fold reports 69.4% accuracy
  - purged K-fold with embargo reports 49.6%

▸ Central chart: Model Comparison: Fold Means

## What connects the steps of the analysis

- Part I: the prediction problem and the basic methods
  - features and fractional differencing, labels, validation without information leakage
- Part II: an S&P 500 case study with formal tests, selection and backtest overfitting
  - machine learning (ML) for inference, a research case study
  - artificial intelligence (AI) for scientific discovery (proofs and estimators: Appendix)
- Abbreviations in the diagram
  - LASSO ($L_1$-penalised regression), RF (random forest), GBM (gradient boosting machine)
  - MLP (multilayer perceptron), CV (cross-validation)
- What you should be able to do after this chapter
  - compute features and labels without look-ahead
  - compute the score of a model validated with purging and embargo
  - derive why truncated FFD of an $I(1)$ price stays $I(1)$
  - interpret, against a benchmark, the Pesaran–Timmermann (PT) and Diebold–Mariano (DM) tests, the deflated Sharpe ratio (DSR), the superior predictive ability (SPA) test and the probability of backtest overfitting (PBO)
  - Further Topics and Appendix: reading for familiarity
- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide
- Most failures come from the process, not the algorithm (López de Prado (2018)); two failures dominate
  - **look-ahead bias**: using information not available at decision time
  - **selection bias**: reporting the best of many trials

## What We Will Explain: from Data to a Tested Claim




- Features use only the information available at the decision date
- Validation measures what still holds on observations the model has not seen
- A predicted direction justifies a trade only after uncertainty and costs are assessed

# Why Machine Learning in Finance, and Why It Is Hard

## The promise

- Flexible, non-linear functional forms
  - interactions found automatically
- Hundreds of candidate predictors (the "factor zoo", Chapter 3)
- Alternative data: text, images, order books, on-chain data

## The constraints

- Very low **signal-to-noise ratio**
  - one serially dependent history that cannot be re-run
- **Non-stationarity**: regimes, breaks
  - markets react to their users
- Labels must be constructed
  - errors cost real money
- Massive risk of **overfitting** (fitting noise)

## Adaptive Markets Hypothesis: Lo (2004)

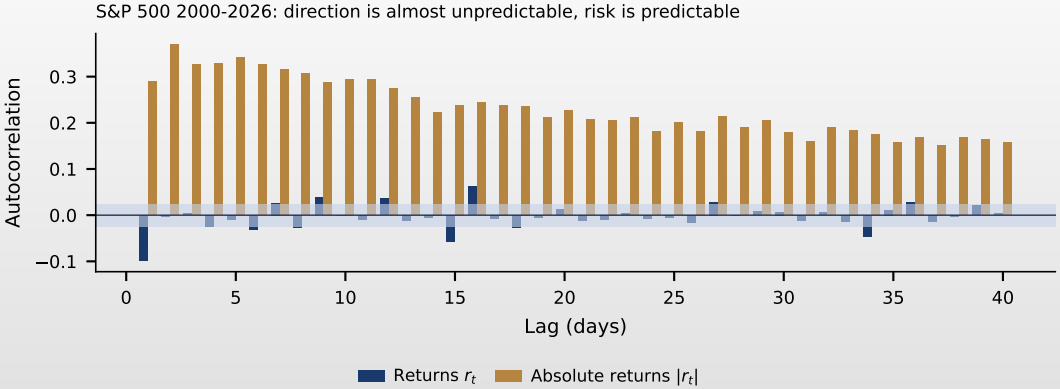
- Predictability appears, is arbitraged away and re-appears
  - a model that worked yesterday is a **hypothesis**, not a law (market efficiency: Chapter 2)
- The hard part is not fitting a model but **not fooling yourself**
  - correct labels, correct validation, honest backtests

# What Is Predictable? Direction vs. Risk (1/2)

- The question: which part of a financial series is predictable at all?
- Autocorrelation at lag $k$: $\rho_k = \mathrm{Corr}(x_t, x_{t-k})$
  - for the daily log return $r_t = \log P_t - \log P_{t-1}$ and for $|r_t|$
- Next chart: lag $k$ on the horizontal axis, $\hat{\rho}_k$ on the vertical axis
  - blue bars for $r_t$, brown bars for $|r_t|$
- The light-blue band, $\pm 1.96/\sqrt{T}$, is the 95% band under no autocorrelation
  - $T$: number of observations
  - for $|r_t|$, the band requires a finite fourth moment, which Chapter 1 questions
- Autocorrelations of returns $r_t$ are smaller than those of $|r_t|$, yet 13 of the 40 lie outside the band
  - at lag 1, $\hat{\rho}_1 = -0.099$
  - the band is neither simultaneous over 40 lags nor robust to heteroskedasticity
  - leaving the band does not show profitability after costs
- Autocorrelations of $|r_t|$ are large and persistent
- ML has the best chance where there is structure
  - **volatility, risk, liquidity, cross-sectional ranking**

MFM_ch13_model_comparison

# What Is Predictable? Direction vs. Risk (2/2)




MFM_ch13_model_comparison ▶ Interactive

# Evidence, and the Economic Value of a Small $R^2_{OOS}$

- Gu, Kelly & Xiu (2020): monthly returns of individual US stocks, 1957–2016
  - they compare linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS)
  - random forests, gradient boosting and neural nets
- Their fit measure: the out-of-sample (OOS) $R^2$
  - how much of the squared return a forecast removes, relative to forecasting zero

$$R^2_{OOS} = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t r_t^2}$$

  - $r_t$: realised return in month $t$ of the test period; $\hat{r}_t$: forecast made with information up to $t-1$
  - the denominator is the error of the zero forecast $\hat{r}_t = 0$: $R^2_{OOS} > 0$ if the model beats it, $R^2_{OOS} < 0$ if it does worse
- Results (Table 1, monthly $R^2_{OOS}$)
  - three-layer network 0.40%, gradient boosting 0.34%, random forest 0.33%
  - OLS with 3 predictors 0.16%, OLS with all 900+ predictors −3.46%
- Trees and nets win through **non-linear interactions**
  - momentum, liquidity and volatility dominate
- The benchmark matters
  - Welch & Goyal (2008) show that most published predictors lose to the **historical mean**
  - Gu, Kelly & Xiu (2020) and Chapter 14 use a zero forecast

## Loss, Penalty and Five Model Families (1/5)

- The question of this section
  - why, with a weak signal, does variance control matter more than flexibility?
- Supervised learning picks, within a model family, the function that fits the training data best
  - after a penalty for complexity

$$\hat{f} = \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} L\big(y_i, f(\mathrm{x}_i)\big) + \lambda\, \Omega(f)$$

  - $(y_i, \mathrm{x}_i)$, $i = 1, \ldots, n$: target and feature vector of observation $i$; $\mathcal{F}$: the model family (e.g. all trees of depth 3)
  - $L$: loss, the squared error $(y - f)^2$ or, for a probability $f$ of $y = 1$, the log-loss $-[y \log f + (1 - y) \log(1 - f)]$
  - $\Omega(f)$: complexity penalty with weight $\lambda \geq 0$; $\lambda = 0$: no penalty, large $\lambda$: a simpler model
- $\lambda$ is chosen by $K$-fold cross-validation (CV) on folds $\mathcal{P}_1, \ldots, \mathcal{P}_K$:

$$CV(\lambda) = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i \in \mathcal{P}_k} L\big(y_i, \hat{f}_\lambda^{(-k)}(\mathrm{x}_i)\big), \qquad \hat{\lambda} = \arg\min_{\lambda} CV(\lambda)$$

  - $n_k$: number of observations in fold $k$; $\hat{f}_\lambda^{(-k)}$: the model fitted on the other $K - 1$ folds

## Loss, Penalty and Five Model Families (2/5)

- Why we penalise: the expected test error at a point $x_0$ (Hastie, Tibshirani & Friedman (2009), Ch. 7)
  - for $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$:

$$\mathbb{E}\big[(y_0 - \hat{f}(x_0))^2\big] = \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\big(\mathbb{E}[\hat{f}(x_0)] - f(x_0)\big)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}\big(\hat{f}(x_0)\big)}_{\text{variance}}$$

  - $\mathbb{E}$: expectation over possible training samples and over the noise of the new observation $y_0 = f(x_0) + \varepsilon_0$
  - bias: the systematic error of a too-simple family
  - variance: the sensitivity of $\hat{f}$ to the training sample; it grows with model flexibility
  - $\sigma^2$: variance of the unpredictable part of $y$, which no model can reduce
- Derivation: add and subtract $\mathbb{E}[\hat{f}(x_0)]$ in $y_0 - \hat{f}(x_0)$
  - the cross terms have mean zero because $\varepsilon_0$ is independent of the training sample
- In finance, return forecasts reduce the error very little
  - an out-of-sample $R^2$ below 1%: the model removes less than 1% of the benchmark's squared loss
  - this does not identify $\sigma^2$: the remaining error also holds misspecification, omitted variables and estimation error
  - so penalties and tree averaging cut variance at the price of a small bias

## Loss, Penalty and Five Model Families (3/5)

- Penalised regressions, with $p$ standardised features and coefficients $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \; \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \mathrm{x}_i^\top \boldsymbol{\beta}\right)^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \tfrac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2\right]$$

  - $\|\boldsymbol{\beta}\|_1 = \sum_j |\beta_j|$, $\|\boldsymbol{\beta}\|_2^2 = \sum_j \beta_j^2$; $\alpha \in [0, 1]$: share of the absolute-value penalty
- Ridge, $\alpha = 0$ (Hoerl & Kennard (1970)): $\hat{\boldsymbol{\beta}} = \left(\mathrm{X}^\top \mathrm{X} + \frac{n\lambda}{2} \mathrm{I}\right)^{-1} \mathrm{X}^\top \mathrm{y}$
  - shrinks all coefficients, none exactly to zero
  - stabilises the estimate when predictors are correlated
- LASSO (least absolute shrinkage and selection operator), $\alpha = 1$ (Tibshirani (1996))
  - the penalty $|\beta_j|$ has a kink at zero, so some coefficients become exactly zero
  - LASSO selects features
- Elastic net, $0 < \alpha < 1$ (Zou & Hastie (2005)): keeps groups of correlated predictors together
  - LASSO, by contrast, arbitrarily keeps one of them
- The chapter's linear benchmark is the penalised logit
  - $p = P(y = 1 \mid \mathrm{x}) = 1/\left(1 + e^{-\mathrm{x}^\top \boldsymbol{\beta}}\right)$, with the log-loss in place of the squared loss

# Loss, Penalty and Five Model Families (4/5)

- Trees: at each node $v$ choose the feature $j$ and threshold $s$ that reduce the Gini impurity the most
  - Gini impurity: $G = 1 - \sum_c \hat{\pi}_c^2$

$$(j^*, s^*) = \arg\min_{j,\, s} \left[ \frac{n_L}{n_v} G(L) + \frac{n_R}{n_v} G(R) \right], \qquad L = \{x_j \le s\},\ R = \{x_j > s\}$$

  - $\hat{\pi}_c$: share of class $c$ in the node; $n_v$, $n_L$, $n_R$: observations in the node and in its two children
  - $G = 0$ for a pure node; with two classes $G = 2\hat{\pi}(1 - \hat{\pi})$, at most 0.5 at 50/50; a leaf predicts the share of class 1
- Random forest (RF, Breiman (2001)): $\hat{f}_{RF}(\mathrm{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathrm{x})$
  - each tree $T_b$ grown on a bootstrap sample with a random subset of features at each split

$$\mathrm{Var}\big(\hat{f}_{RF}(\mathrm{x})\big) = \rho\, \sigma_T^2 + \frac{1 - \rho}{B}\, \sigma_T^2$$

  - $\sigma_T^2$: variance of one tree; $\rho$: correlation between two trees (Hastie, Tibshirani & Friedman (2009), Ch. 15)
  - random feature subsets lower $\rho$, the only term that more trees cannot remove
  - the out-of-bag error is **optimistic** when labels overlap

Random Forests

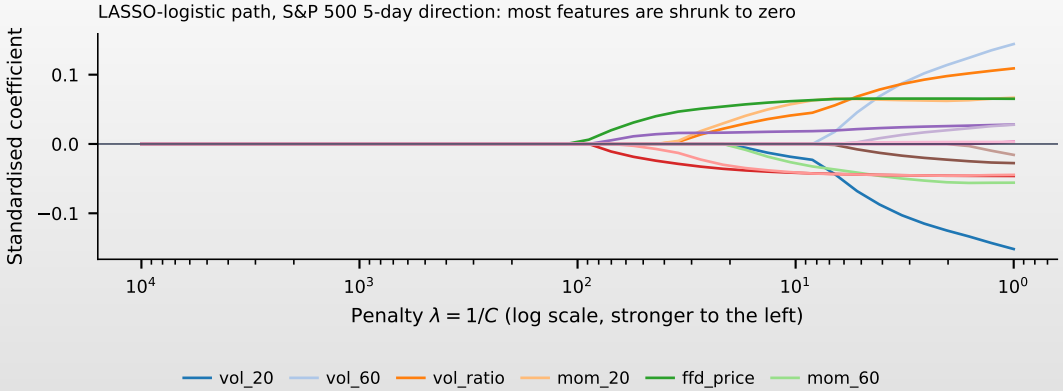
# Loss, Penalty and Five Model Families (5/5)

- Gradient boosting machine (GBM, Friedman (2001))
  - add, step by step, a small tree that corrects the errors of the current model

$$r_{im} = -\frac{\partial L(y_i, F)}{\partial F}\bigg|_{F=F_{m-\mathbf{1}}(\mathbf{x}_i)}, \qquad F_m(\mathbf{x}) = F_{m-\mathbf{1}}(\mathbf{x}) + \eta\, h_m(\mathbf{x})$$

  - $F_m$: the model after $m$ steps; $h_m$: the tree fitted to the pseudo-residuals $r_{im}$; for the log-loss on the logit score, $r_{im} = y_i - \hat{p}_{m-\mathbf{1}}(\mathbf{x}_i)$
  - $\eta \in (0, 1]$: learning rate; a small $\eta$ learns slowly and overfits less
- Multilayer perceptron (MLP): hidden layers $\mathbf{h}^{(\ell)} = g\big(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-\mathbf{1})} + \mathbf{b}^{(\ell)}\big)$, $\mathbf{h}^{(\mathbf{0})} = \mathbf{x}$
  - the output: $\hat{p} = 1/\big(1 + e^{-(\mathbf{w}^\top \mathbf{h}^{(D)} + b)}\big)$
  - $\mathbf{W}^{(\ell)}$, $\mathbf{b}^{(\ell)}$: weights and biases of layer $\ell$; $D$: number of hidden layers; $g(z) = \max(0, z)$: the ReLU (rectified linear unit), element by element
  - the weights minimise the log-loss by gradient descent
  - early stopping: a chronological validation set (the last 10% of the training data, purged); training stops when the validation log-loss stops falling
  - richer architectures are in Chapter 14

# LASSO Regularisation Path (1/2)




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# LASSO Regularisation Path (2/2)

- $L_1$-penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Chart labels
  - vol_20, vol_60, vol_ratio: the 20- and 60-day volatilities and their ratio
  - mom_20, mom_60: the log returns over the last 20 and 60 days
  - ffd_price: the FFD log price
- Horizontal axis: the penalty $\lambda = 1/C$, on a log scale, stronger to the left
  - vertical axis: the standardised coefficient
- Strong penalty (left of the chart): every coefficient is zero
  - the first feature to enter is the FFD (fixed-width window fractional differentiation) log price
  - FFD log price: a scaled price level (next section)
- Weak penalty: the volatility features get the largest coefficients
- All coefficients are small in absolute value
  - weak signal, consistent with near-efficient markets

MFM_ch13_regularization

## Stationarity, Memory and the Fractional Difference (1/2)

- The question of this section
  - how do we turn a price into a stationary feature without erasing its memory?
- Log prices $X_t = \log P_t$ behave as $I(1)$
  - $P_t$: closing price on day $t$
- Returns $\Delta \log P_t = \log P_t - \log P_{t-1}$ are $I(0)$ but **erase the memory** of the price level
- Integration order: a series $X_t$ is $I(\delta)$ if $(1-B)^\delta X_t$ is $I(0)$
  - $B$: backshift operator, $BX_t = X_{t-1}$
  - $I(0)$: stationary with short memory; $I(1)$: a random walk type series, stationary after one difference
  - $\delta$ may be fractional: the series is stationary only for $\delta < \frac{1}{2}$, with long memory when $0 < \delta < \frac{1}{2}$ (Hosking (1981))
- Idea of fractional differencing: difference by less than 1
  - removing the unit root while keeping part of the memory

# Stationarity, Memory and the Fractional Difference (2/2)

## Definition 1 (Fractional difference of order $d$)

- $\tilde{X}_t = (1-B)^d X_t = \sum_{k=0}^{\infty} w_k X_{t-k}$, with $w_0 = 1$ and $w_k = -w_{k-1}\frac{d-k+1}{k}$

- Notation
  - $d$: differencing order chosen by the analyst; $k$: lag in days
  - $w_k$: weight of the value $k$ days back, from the binomial expansion of $(1-B)^d$
- $d = 1$: $w = (1, -1, 0, \dots)$, ordinary returns
- $0 < d < 1$: weights decay hyperbolically, so memory is kept
- An $I(1)$ log price differenced with $d$ is $I(1-d)$
  - non-stationary for $d \leq \frac{1}{2}$
  - stationary with long memory for $\frac{1}{2} < d < 1$
- The infinite weights sum to $(1-1)^d = 0$
  - the level cancels only in the limit (asymptotics in the Appendix)
- López de Prado (2018) proposes the minimum $d$ at which an ADF (Augmented Dickey–Fuller) unit-root test rejects
  - test equation and critical value on the ADF slide; this section shows why that rule fails

## The Fixed-Width Window (FFD)

**Definition 2 (Fixed-width-window fractional differentiation (FFD))**

- Truncate the weights: keep $w_0, \ldots, w_K$ with $|w_k| \geq \tau$ (e.g. $\tau = 10^{-4}$), drop the rest (López de Prado (2018))
- $\tilde{X}_t = \sum_{k=0}^{K} w_k X_{t-k}$ for every $t > K$

- $\tau$: smallest weight kept; $K$: the last lag kept, i.e. the window length
  - the first $K$ days have too little history and are dropped
- Every $\tilde{X}_t$ uses the same number of lags
  - only past values enter $\Rightarrow$ no look-ahead
- The smaller $d$, the more lags are needed
  - $d = 0.25$, $\tau = 10^{-4} \Rightarrow K = 444$ lags
- The truncated weights no longer sum to zero
  - $\sum_{k=0}^{K} w_k = 0.178$ at $d = 0.25$ (0.040 at $d = 0.5$)

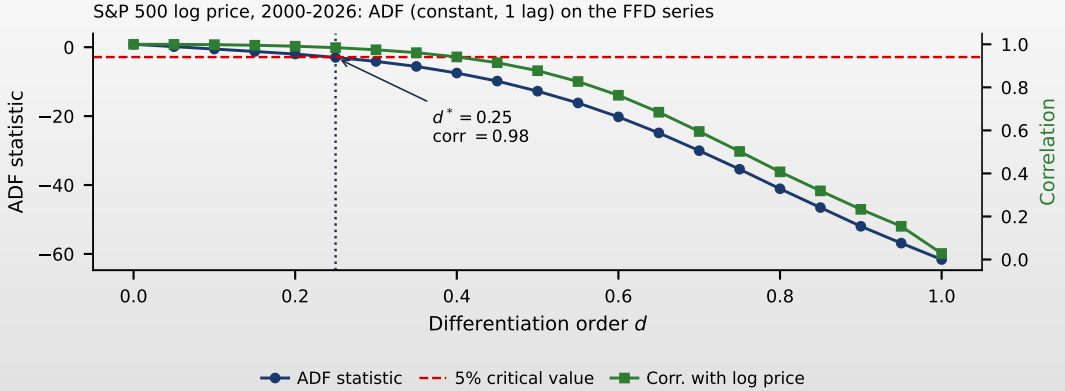
# The ADF Statistic Along $d$: S&P 500 (1/3)

- Augmented Dickey–Fuller (ADF) unit-root test: the $t$-statistic of $\gamma$ in

$$\Delta \tilde{X}_t = c + \gamma \, \tilde{X}_{t-1} + \sum_{i=1}^{p} \phi_i \, \Delta \tilde{X}_{t-i} + e_t$$

  - $H_0$: $\gamma = 0$, a unit root; $p$: number of lagged differences, which absorb the autocorrelation of $e_t$
  - with a constant, reject $H_0$ at 5% below $\approx -2.86$
- The rule of López de Prado (2018)
  - compute the FFD series on a grid of $d$ between 0 and 1
  - pick the smallest $d$ at which the ADF rejects
- Next chart: $d$ on the horizontal axis
  - the ADF statistic (blue) on the left axis
  - the correlation of the FFD series with the log price (green) on the right axis
  - red dashed line: the 5% critical value; vertical line: $d^* = 0.25$

# The ADF Statistic Along $d$: S&P 500 (2/3)




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# The ADF Statistic Along $d$: S&P 500 (3/3)

- FFD log price, S&P 500 2000–2026, constant and 1 lag: ADF $= -2.96$
  - so the rule rejects from $d = 0.25$
- At $d = 0.25$, $\mathrm{Corr}(\tilde{X}_t, \log P_t) = 0.98$
  - the FFD series is almost the log price
- Same series, with 34 lags chosen by the Akaike information criterion (AIC)
  - ADF $= -0.72$, no rejection
- The verdict depends on the number of lags
- A correlation of 0.98 with an $I(1)$ price is a warning, not a virtue
- What the result supports: comparison of procedures
  - the same log price, two lag choices, two verdicts

MFM_ch13_ffd

# Truncated FFD of an $I(1)$ Price Is Still $I(1)$

- Decompose the FFD series:

$$\tilde{X}_t = \sum_{k=0}^{K} w_k X_{t-k} = \Big(\sum_{k=0}^{K} w_k\Big) X_t - \sum_{k=1}^{K} w_k \left(X_t - X_{t-k}\right)$$

  - first term: today's level times the sum of the truncated weights; second term: a weighted sum of the $k$-day changes $X_t - X_{t-k}$
- If $X_t$ is $I(1)$, each $X_t - X_{t-k}$ is $I(0)$
  - the first term is $0.178\,X_t$ at $d = 0.25$: a **scaled random walk**
- The unit-root component has small variance next to the stationary part
  - so ADF with few lags confuses the two
  - Monte Carlo: pure random walk, $T = 6{,}274$ days, FFD with $d = 0.25$: ADF (1 lag) rejects in **97.8%** of 500 samples, at a nominal 5%
- A finite filter whose weights do not sum to zero keeps a level component, whatever the ADF p-value
- Consequence: the FFD log-price in the case study is a scaled price level, not a stationary feature
  - the memory is measured directly in Further Topics

MFM_ch13_inference

## From Fixed-Horizon to Triple-Barrier Labels (1/2)

- The question of this section
  - which label reproduces a trader's real decision, with take-profit, stop-loss and a maximum holding period?
- **Fixed horizon**: the label is the sign of the $h$-day log return
  - with a dead zone around zero

$$r_{t,t+h} = \log P_{t+h} - \log P_t, \qquad y_t = \begin{cases} +1 & \text{if } r_{t,t+h} > \tau \\ -1 & \text{if } r_{t,t+h} < -\tau \\ 0 & \text{otherwise} \end{cases}$$

  - $P_t$: closing price on day $t$; $h$: horizon in days; $\tau \geq 0$: threshold, moves smaller than $\tau$ get the label 0
- Three problems
  - a constant $\tau$ ignores time-varying volatility
  - the path is ignored: a stop-loss could have been hit before $t + h$
  - the classes are unbalanced: S&P 500, $h = 10$, $\tau = 0$: 60.5% up vs. 39.5% down

# From Fixed-Horizon to Triple-Barrier Labels (2/2)

## Definition 3 (Triple-barrier label, López de Prado (2018))

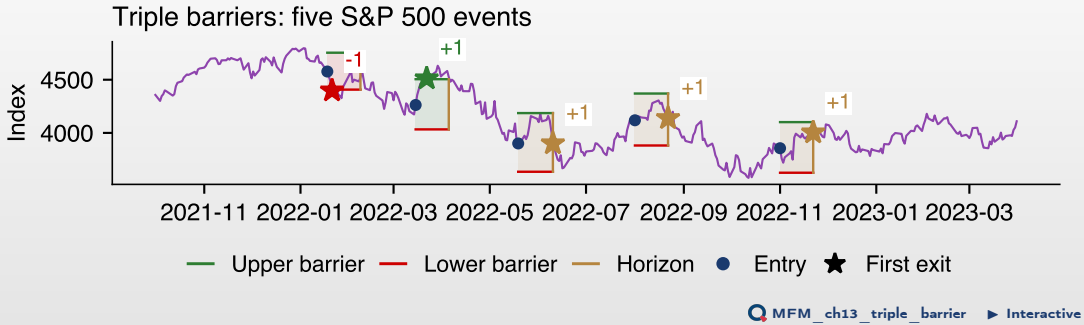
- At event time $t_0$, with daily volatility $\sigma_{t_0}$ and horizon $h$
  - upper barrier $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$ (profit-taking)
  - lower barrier $\leq -\pi_l \sigma_{t_0} \sqrt{h}$ (stop-loss), vertical barrier $t = t_0 + h$
- Label = **first barrier touched** at $t_1$
  - $+1$ upper, $-1$ lower; at the vertical barrier $\text{sign}(r_{t_0,t_1})$ (or 0)

- $\sigma_{t_0} \sqrt{h}$: the typical $h$-day move; $\pi_u, \pi_l > 0$: barrier widths in units of it
- Each label has a **span** $[t_0, t_1]$: the input of purging in validation
- $\sigma_{t_0}$: exponentially weighted moving average (EWMA) volatility, span 50
  - square root of the EWMA of squared deviations of daily log returns

$$\sigma_t^2 \approx \alpha \sum_{k \geq 0} (1-\alpha)^k \left(r_{t-k} - \bar{r}_t\right)^2, \qquad \alpha = \frac{2}{\text{span} + 1} = \frac{2}{51} \approx 0.039$$

  - $\bar{r}_t$: the exponentially weighted mean return; span 50: an effective memory of about two trading months
- Why volatility-scaled barriers
  - a fixed threshold would be hit almost daily in 2008 or March 2020 and rarely in calm years

## Triple Barrier: S&P 500 in 2022 (1/2)

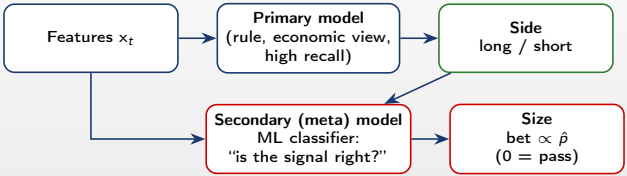


## Triple Barrier: S&P 500 in 2022 (2/2)

- Five events during the 2022 bear market, $\pi_u = \pi_l = 1$, $h = 15$ days
  - the colour of each star is the barrier touched first
- Chart: the index level
  - green lines: upper barrier, red: lower barrier, brown: horizon
  - dots: entry, stars: first barrier touched
- January: stop-loss hit ($-1$); March: profit-taking ($+1$)
  - the other three end at the vertical barrier (amber stars) above their entry level
  - so the sign rule labels them $+1$
- A label $+1$ therefore does not mean that the profit barrier was touched

MFM_ch13_triple_barrier

# Meta-Labelling: Deciding Whether to Act (1/2)




- Meta-label $y_t^{meta} = 1$ if the primary signal was profitable, 0 otherwise
  - profitable: barrier touched in its direction
- The secondary model estimates $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$
  - the probability that the signal of day $t$ is right given the features $x_t$
- ML learns **when to trust** an interpretable primary rule, not the direction itself

## Meta-Labelling: Deciding Whether to Act (2/2)

- A filter decides which signals to act on; three scores describe it

$$\text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

  - TP: profitable signals taken; FP: unprofitable signals taken; FN: profitable signals skipped
  - acting on every signal gives recall 1; $F_1$: their harmonic mean, between 0 and 1
- Example (Seminar 13, exercise B11): moving-average crossover on Bitcoin
  - the 20- and 50-day moving-average crossover (MA20/MA50), 1,660 test trades
  - with the meta filter, precision rises from 0.475 to 0.517, recall falls from 1 to 0.541, and $F_1$ falls from 0.644 to 0.529
  - the mean gross log return per trade, before costs, moves from $-0.27\%$ to $+0.19\%$
- Meta-labelling raises precision at the cost of recall
  - with costs per trade, precision and the return per trade matter, not $F_1$

## Leakage: What Crosses the Train/Test Boundary

- The question of this section
  - why does ordinary $K$-fold overstate a model's skill on financial series?

**Definition 4 (Leakage)**

- The training set contains information that is also present in the test set
  - so the test score no longer measures performance on genuinely new data

- Serially correlated features ($x_t \approx x_{t+1}$) and overlapping labels ($y_t \approx y_{t+1}$)
  - a shuffled $K$-fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour $\Rightarrow$ excellent "out-of-sample" score
  - tuning on a leaky CV selects the **most overfitted** model
- Other sources of leakage
  - transformations fitted on the full sample (scaling, principal component analysis, the choice of $d$ for FFD)
  - survivorship bias, data revisions, wrong timestamps

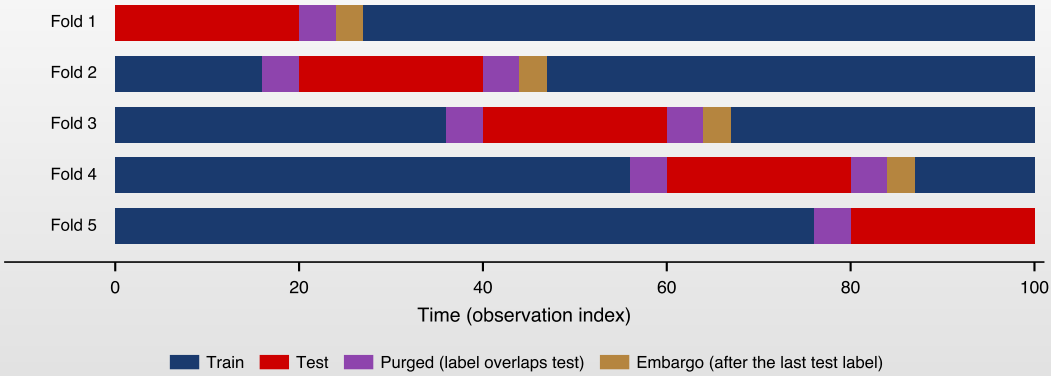
# Purged K-Fold with Embargo (1/2)

- Folds are **contiguous** blocks in time (López de Prado (2018))
  - test fold $k$ holds the observations $t_k^a, \ldots, t_k^b$
  - their labels end at the latest at $T_k = \max_{i \in \text{test}_k} t_{1,i}$
- The training set keeps only the observations that do not touch the test:

$$\mathcal{A}_k = \left\{ i \notin \text{test}_k : \; [t_{0,i}, t_{1,i}] \cap [t_k^a, T_k] = \emptyset \;\; \text{and} \;\; t_{0,i} \notin (T_k, \; T_k + e] \right\}$$

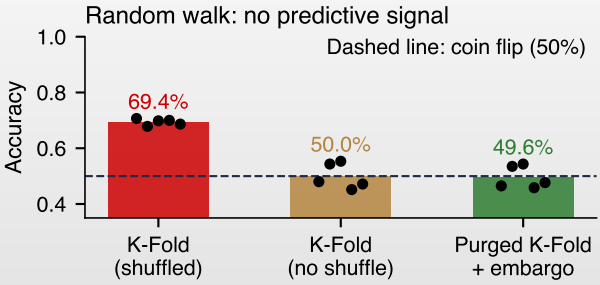
  - the first condition is **purging**: it removes labels that overlap the span of the test labels
  - the second is the **embargo**: $e$ observations after the end of the last test label, $T_k$ (e.g. 1%), needed because features look back in time
- The purged CV score is the mean of $K$ scores
  - each from a model fitted on $\mathcal{A}_k$ and evaluated on fold $k$
- With labels of length $h$, a boundary removes $h$ observations before and $h + e$ after the test block
  - an interior fold loses $2h + e$ observations
  - a 10-observation example: Appendix
- Next chart: five folds in rows, observation index on the horizontal axis
  - blue train, red test, purple purged, brown embargo

# Purged K-Fold with Embargo (2/2)




MFM_ch13_purged_cv ▶ Interactive

# Leakage Experiment with an Unpredictable Target




- Price = pure random walk
  - label = sign of the 20-day forward return
  - 5 features = 20-day rolling means of independent noise
- True accuracy is 50% by construction
- Random forest, 5 folds
- **Shuffled K-fold: 69.4%**
  - contiguous K-fold 50.0%
  - purged K-fold + embargo 49.6%
- Shuffling fails because labels overlap *and* features persist
  - here contiguous folds already remove most of the leak: only the fold boundaries share returns, and purging cleans them
  - a result that looks too good is usually leakage
- What the result supports: comparison of procedures

MFM_ch13_purged_cv ▶ Interactive

## Model Selection versus Deployment

- Ordinary $K$-fold is valid for a purely autoregressive model with serially uncorrelated errors (Bergmeir, Hyndman & Koo (2018))
  - when it fails: Appendix
- **Model selection**: purged $K$-fold, or combinatorial purged CV (CPCV)
  - CPCV tests on many combinations of groups and returns a **distribution** of Sharpe ratios instead of one number
- **Deployment**: walk-forward (WF)
  - train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward ($T_k$: end of the $k$-th training window)
  - realistic, but a single path with high variance: it serves the final evaluation, not the choice (WF and CPCV: Appendix)

## Case Study: Can the 5-Day Direction of the S&P 500 Be Predicted?

- S&P 500 index and VIX, daily closes of the course data, 2001–2026: $n = 6{,}269$ observations
  - VIX: the CBOE volatility index, the 30-day implied volatility of S&P 500 options
- Why the S&P 500: the most liquid equity index, with a long VIX history
  - if its direction were predictable, arbitrage would be cheap
- Target: $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$
  - equal to 1 if the index is higher 5 days later
  - the fixed-horizon label with $h = 5$, $\tau = 0$
- Why 5 days: a weekly trading horizon
  - about $6{,}269/5 \approx 1{,}250$ non-overlapping labels
- The label span $[t, t + 5]$ is used for purging
- Validation: purged 5-fold CV, 1% embargo, the same folds for every model
  - each fold covers about 5 years, hence at least one crisis episode

## Case Study: Metrics, against the "Always Up" Baseline

- Accuracy: share of days whose direction is predicted correctly
- Receiver operating characteristic (ROC) area under the curve (AUC)
  - the probability that a randomly chosen up day gets a higher predicted probability than a randomly chosen down day
- AUC $= 0.5$: no skill; AUC $= 1$: perfect ranking
- Example: up days with probabilities 0.7 and 0.4, down days with 0.5 and 0.2
  - 3 of the 4 pairs are ranked correctly, so AUC $= 0.75$
- Why both metrics
  - accuracy depends on the 0.5 threshold
  - AUC compares only the ranking of the probabilities
- **Fold mean**: the metric computed in each of the 5 test folds, then averaged
- **Pooled**: the metric computed once on all out-of-fold predictions together
  - used by the formal tests

# Case Study: 14 Features and Four Models

## Features, information up to day $t$

- **Momentum**: $\log P_t - \log P_{t-h}$
  - the log return over the last $h$ days, $h \in \{1, 5, 20, 60, 120\}$
- **Realised volatility**
  - annualised standard deviation (SD) of daily log returns over 20 and 60 days, and their ratio
- **Distance to moving average**
  - $\log P_t$ minus the mean of $\log P$ over 50 or 200 days
- **RSI(14)**: $100 - 100/(1 + RS)$
  - $RS$ = smoothed average gain over average loss, 14 days
  - between 0 and 100, high after rises
- **VIX**: log level and 5-day change
- **FFD** log price, $d = 0.25$
  - chosen on the full sample (ablation after the model comparison)

## Models, deliberately conservative

- **Logit**: L2-penalised (Ridge), standardised features
- **RF**: 400 trees, minimum leaf 100
- **GBM**: histogram-based
  - depth 3, learning rate 0.03, 200 trees
- **MLP**: hidden layers of 32 and 16 units, early stopping on a purged chronological validation set
- Large leaves and shallow trees limit variance

# A Classifier That Predicts Only the Majority Class (1/2)

| | **Pred. up** | **Pred. down** |
|---|---|---|
| **Actual up** | TP = 3,575 | FN = 65 |
| **Actual down** | FP = 2,589 | TN = 40 |

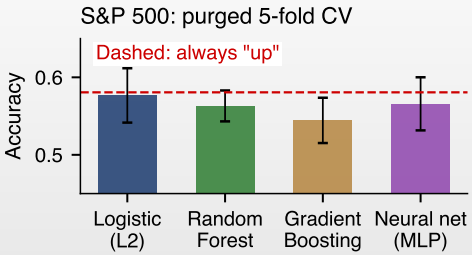
- L2-penalised logit, $n = 6{,}269$
  - pooled out-of-fold predictions of purged 5-fold CV
- TP, FN, FP, TN: true positives, false negatives, false positives, true negatives

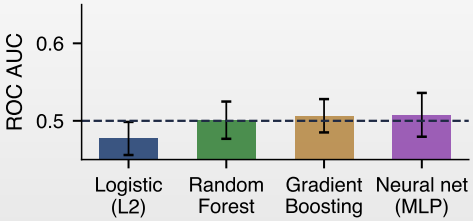
- Accuracy $= (TP + TN)/n = 57.7\%$
- Precision $= TP/(TP + FP) = 0.580$
- Recall $= TP/(TP + FN) = 0.982$
- $F_1 = 0.729$ (formula on the meta-labelling slide)
- The model predicts "up" on 98.3% of days
  - it has essentially learned "always up"

# A Classifier That Predicts Only the Majority Class (2/2)

- **Majority baseline** from the same table
  - actual up days $(3{,}575 + 65)/6{,}269 = 3{,}640/6{,}269 = 58.1\%$
  - the accuracy of predicting "up" every day, with no model at all
- A 57.7%-accurate model is *worse* than doing nothing
- The cause is class imbalance
  - with weak features the predicted probabilities stay near the base rate 0.58, above the 0.5 threshold
  - so almost every day is classified "up"
  - the case study keeps the fixed-horizon label on purpose, so that the baseline is transparent
  - triple-barrier labels change the class balance the classifier sees (Seminar 13, exercises B1 and B8)
- Other baselines: yesterday's direction, a moving-average rule, buy-and-hold for trading
- A model has value only if it beats the baseline **out of sample**
  - after costs, by more than sampling error

# Model Comparison: Fold Means




| Fold mean | Logit | RF | GBM | MLP |
|---|---|---|---|---|
| Accuracy | 57.7% | 56.3% | 54.4% | 56.6% |
| AUC | 0.477 | 0.501 | 0.507 | 0.508 |

- Bars: mean over the 5 purged folds; whiskers: $\pm 1$ SD across folds
- Baseline "always up": **58.1%**; no model beats it

MFM_ch13_model_comparison ▶ Interactive



▸ Further Topics: FFD ablation, Newey–West

# Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (1/2)

- **Pesaran–Timmermann** (PT, Pesaran & Timmermann (1992)): are predicted and realised directions related?

$$PT = \frac{\hat{P} - \hat{P}^*}{\widehat{\mathrm{SE}}} \xrightarrow{d} N(0,1), \qquad \hat{P}^* = \hat{p}_y \hat{p}_x + (1-\hat{p}_y)(1-\hat{p}_x)$$

  - $\hat{P}$: share of days with the correct sign; $\hat{p}_y$, $\hat{p}_x$: shares of realised and of predicted "up"; $\hat{P}^*$: share correct expected if predictions were independent of outcomes
  - $H_0$: independence; $\widehat{\mathrm{SE}}$: standard error of $\hat{P} - \hat{P}^*$ under $H_0$ (Appendix)
  - $PT > 1.645$: skill at 5% (one-sided); $PT < 0$: fewer hits than chance
- PT assumes independent days
  - with overlapping labels we also use the slope $t$ in $y_t = a + b\,x_t + u_t$
  - Newey–West errors with $q = 4$ (Newey–West slide)
  - $x_t$ the predicted and $y_t$ the realised direction, $b > 0$ means skill
- **AUC**: the DeLong standard error (SE) assumes independent days (Appendix)
- Hence a circular block bootstrap on blocks of $L = 20$ consecutive days (Künsch (1989))
  - four times the label length, which keeps the serial dependence

# Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (2/2)

| Pooled out-of-fold | Logit | RF | GBM | MLP |
|---|---|---|---|---|
| $\hat{P}$ / $\hat{P}^*$ | 0.577 / 0.578 | 0.563 / 0.570 | 0.544 / 0.554 | 0.566 / 0.569 |
| $PT$ / HAC $t$ (lag 4) | −0.80 / −0.62 | −2.27 / −1.67 | −1.97 / −1.49 | −0.83 / −0.60 |
| AUC | 0.480 | 0.495 | 0.501 | 0.497 |
| block-bootstrap 95% CI | [0.456; 0.505] | [0.470; 0.520] | [0.478; 0.525] | [0.474; 0.520] |

- No significant positive directional skill: every block-bootstrap AUC interval contains 0.5
- At HAC lag 4, no model differs significantly from chance ($|t| \leq 1.67$)
  - PT, which treats the days as independent, gives −2.27 for RF: overlapping labels inflate significance
- Logit, for contrast: the DeLong 95% CI is [0.466; 0.494] and excludes 0.5
  - the DeLong CI treats days as independent
- Consistent with weak-form efficiency (Chapter 2)
- What the result supports: comparison of procedures
- Look for signal in cross-sections (Gu, Kelly & Xiu (2020))
  - in risk targets such as volatility (Chapters 5 and 9)
  - in crypto delisting (Będowska-Sójka, Wójcik & Pele (2026)) and in text (Chapter 15)

MFM_ch13_inference

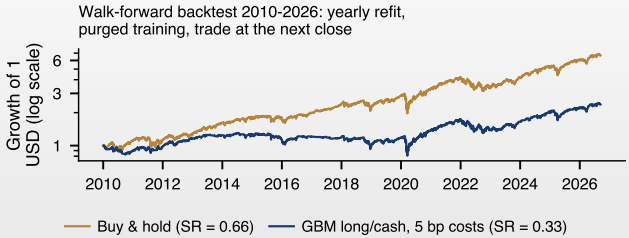
# Walk-Forward Backtest: Strategy and Design

- The question
  - would a trader following the model have earned more, after costs, than by simply holding the index?
- Position $w_t \in \{0, 1\}$: long the S&P 500 if $\hat{p}_t > 0.5$, otherwise cash
  - decided with information up to $t$
  - $\hat{p}_t$: the model's probability of "up"
- Strategy return: the index return when invested
  - minus a cost each time the position changes
  - the decision at the close of day $t$ is traded at the close of day $t + 1$

$$r^{s}_{t+\mathbf{2}} = w_t \, r_{t+\mathbf{2}} - c \, |w_t - w_{t-\mathbf{1}}|$$

  - $r_{t+\mathbf{2}}$: log return of the index from the close of $t + 1$ to that of $t + 2$; turnover $|w_t - w_{t-\mathbf{1}}|$: 1 on the days the position changes
  - $c = 5$ basis points (bp), i.e. 0.05% per unit traded: a prudent cost for an exchange-traded fund on the S&P 500
  - $r - c$ approximates $\log\left((1 - c)\, e^{r}\right) = r + \log(1 - c)$, the log return after paying the cost
- Exposure $\bar{w} = \frac{\mathbf{1}}{T} \sum_t w_t$: share of the $T$ days spent invested
- Cash earns 0; interest on cash days is ignored
  - both Sharpe ratios use daily log returns with risk-free rate 0, annualised with $\sqrt{252}$
- Model: GBM refitted every January, 2010–2026, on all past data
  - purging labels that overlap the test year; FFD log price with $d = 0.15$, chosen on data up to 2009
- Benchmark: buy-and-hold ($w_t = 1$: exposure 100%, no turnover)

# Walk-Forward Backtest: Results (1/2)




- 2010–2026: strategy Sharpe **0.33** vs. buy-and-hold **0.66** (annualised)
  - average exposure 76%
- The model stayed mostly long
  - its exits from the market (24% of the days) lowered the Sharpe ratio
- Chart: growth of one dollar invested, log scale
  - brown buy-and-hold, blue the GBM strategy

MFM_ch13_walk_forward ▶ Interactive

## Walk-Forward Backtest: Results (2/2)

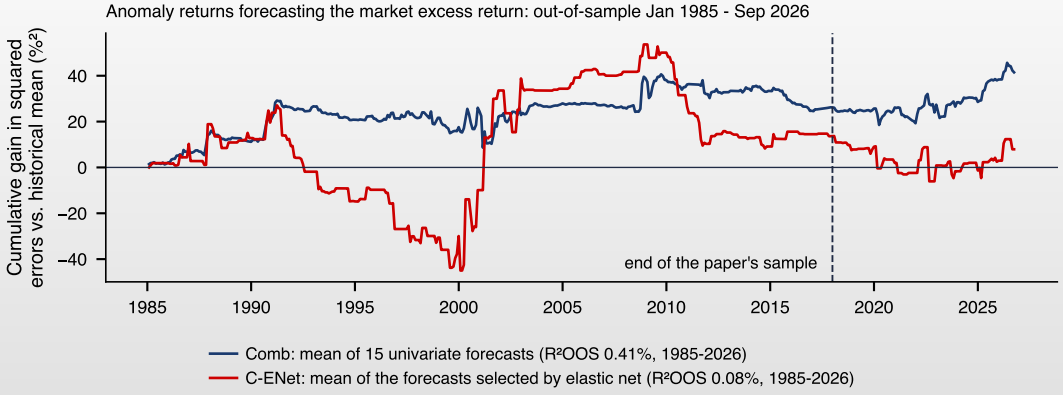
- The Sharpe-ratio difference, $-0.33$, has $p = 0.001$
  - Ledoit–Wolf HAC test, in Further Topics
  - requires a finite fourth moment, which Chapter 1 questions
- The strategy is significantly worse than buy-and-hold; there is no evidence of skill
- What the result supports: comparison of procedures (the GBM strategy against buy-and-hold)
- Sensitivity of the strategy SR
  - 0.31 with $d = 0.25$, chosen on the full sample, hence with information from the test period
  - 0.48 without the FFD log price
  - 0.43 when trading at the same close, an idealisation with no execution delay

MFM_ch13_walk_forward

# Case Study: Do Cross-Sectional Anomalies Predict the Market?

- Dong, Li, Rapach & Zhou (2022): do long-short anomaly returns predict next month's market excess return?
  - anomaly: a portfolio long the stocks with high average returns on a characteristic and short those with low average returns
  - examples: value (the book-to-market ratio), momentum, accruals, net share issuance
- The cross-sectional literature and market-return predictability had been studied separately
- The benchmark is hard to beat: Welch & Goyal (2008) show that most published predictors do not beat the historical mean
- The paper's data: 100 US anomalies, monthly returns, January 1970 – December 2017
  - the first out-of-sample forecast has 180 months for 100 predictors
  - the regression on all predictors learns noise
- Method: shrinkage, i.e. elastic net, forecast combination and dimension reduction
- Main claim: anomalies have statistically and economically significant out-of-sample predictive power for the market
- Proposed explanation: asymmetric limits of arbitrage and persistent correction of overpricing
  - overpricing is harder to arbitrage than underpricing, because short selling is costly
  - hence the short leg of the anomalies carries most of the information

# Anomalies and the Market: Out-of-Sample Gain over the Historical Mean




MFM_ch13_anomaly_market ▶ Interactive

# Anomalies and the Market: Method and What the Replication Does Differently

- Predictive regression of anomaly $i$, estimated on an expanding window and re-estimated every month

$$r_{t+\mathbf{1}} = a_i + b_i\, x_{i,t} + \varepsilon_{i,t+\mathbf{1}}, \qquad \hat{r}^{(i)}_{t+\mathbf{1}} = \hat{a}_i + \hat{b}_i\, x_{i,t}$$

  - $r_{t+\mathbf{1}}$: market excess return (Mkt-RF) in month $t+1$
  - $x_{i,t}$: long-short return of anomaly $i$ in month $t$
- Comb (simple combination): the mean of the $N$ forecasts $\hat{r}^{(i)}_{t+\mathbf{1}}$
- C-ENet (combination elastic net): elastic net in the regression of $r_t$ on the $N$ forecasts, with non-negative coefficients
  - the forecast is the mean of the selected forecasts
  - ENet: elastic net on all predictors; OLS: the regression on all predictors
- The paper's periods: training 1970–1979, validation 1980–1984, out-of-sample 1985–2017
- The replication, on the Kenneth French Data Library
  - 15 anomalies instead of 100: SMB, HML, RMW, CMA (Fama & French (2015)), momentum, short- and long-term reversal
  - and the extreme-decile spreads for accruals, net issuance, beta, variance, residual variance, E/P, CF/P, D/P
  - elastic-net penalty ($\ell_{\mathbf{1}}$ and $\ell_{\mathbf{2}}$ in equal proportions) chosen by the small-sample corrected AIC
  - out-of-sample period extended to September 2026

# Anomalies and the Market: Result and Limits

- Measure: $R^2_{OOS}$ against the historical mean $\bar{r}_t$, computed up to month $t-1$ (Campbell & Thompson (2008))

$$R^2_{OOS} = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t (r_t - \bar{r}_t)^2}$$

  - the chart cumulates the squared errors of the historical mean minus those of the model
  - the curve rises in the months in which the model beats the historical mean
- The paper's figures, published version (Table II), 1985–2017: C-ENet 2.81%, ENet 2.03%, Comb 0.89%, OLS −2,513.86%
- The replication, 1985–2017: Comb 0.35%, C-ENet 0.18%, ENet −4.75%, OLS −6.84%
- The replication, 1985–2026: Comb 0.41% (one-sided Clark–West $p$ 0.045), C-ENet 0.08% ($p = 0.15$)
  - Clark–West: the adjusted difference regressed on a constant, Newey–West errors with $\lfloor 4(T/100)^{2/9} \rfloor = 5$ lags
  - Comb $p$ with 3, 6 and 12 lags: 0.063, 0.044 and 0.032; without the HAC correction: 0.068
  - the lag count is fixed by the rule, not chosen by significance
  - the Clark & West (2007) test requires a finite fourth moment, which Chapter 1 questions
- After the paper's sample, 2018–2026: Comb 0.61% ($p = 0.12$), C-ENet −0.23%
- What the result supports: the ranking of procedures (forecast combination ahead of the regression on all predictors)
  - it does not support the size of the paper's gains
- Limits: 15 anomalies do not reproduce the 100
  - with 15 forecasts, C-ENet selects no forecast in 57% of the months and then uses the historical mean

MFM_ch13_anomaly_market

## Selection Bias and the False Strategy Theorem

- The question of this section
  - how large is the best Sharpe ratio among $N$ trials when none has skill?
- A researcher tries $N$ strategies and reports the best: **selection bias**
  - even if all have zero true Sharpe ratio, $\max_n \widehat{SR}_n$ grows with $N$
- Harvey, Liu & Zhu (2016): given hundreds of published factors, a new factor needs $t > 3$, not 2

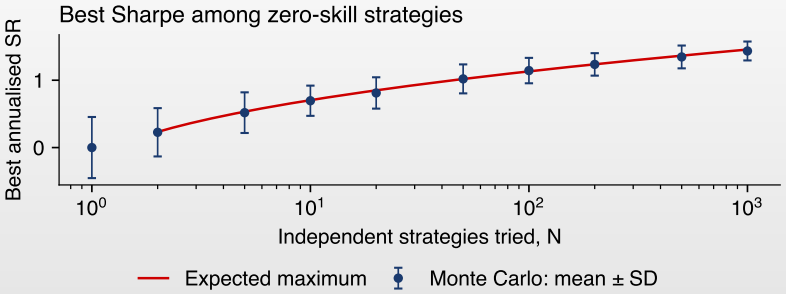
### Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- The **expected maximum Sharpe ratio** of $N$ strategies with no skill
  - if $\widehat{SR}_n$, $n = 1, \ldots, N$, are independent and approximately $N(0, \mathrm{Var}[\widehat{SR}])$, then

$$SR_0 \equiv \mathbb{E}\Big[\max_n \widehat{SR}_n\Big] \approx \sqrt{\mathrm{Var}[\widehat{SR}]}\left[(1-\gamma_E)\,\Phi^{-1}\Big(1-\tfrac{1}{N}\Big) + \gamma_E\,\Phi^{-1}\Big(1-\tfrac{1}{Ne}\Big)\right]$$

- Notation
  - $\widehat{SR}_n$: estimated Sharpe ratio of strategy $n$; $\Phi$: standard Normal distribution function; $\Phi^{-1}$: its quantile function
  - $\mathrm{Var}[\widehat{SR}]$: sampling variance of one estimated SR, $\approx 1/T$ per period for i.i.d. returns, $T$ the number of return periods
  - $\gamma_E \approx 0.5772$: Euler–Mascheroni constant
- $SR_0$ is the Sharpe ratio the luckiest of $N$ worthless strategies is expected to show
  - it rises with $N$ and with the sampling noise $\sqrt{\mathrm{Var}[\widehat{SR}]}$

# The Best of $N$ Zero-Skill Strategies




- Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised $\mathrm{Var}[\widehat{SR}] \approx 1/5$)
- Trying $N = 1000$ variants yields a best Sharpe ratio of $\approx 1.45$ — **with no skill at all**
- Horizontal axis: number of strategies $N$ (log scale); vertical axis: best annualised SR
  - red line: the theorem; dots: Monte Carlo mean $\pm$ one SD, in line with the theorem

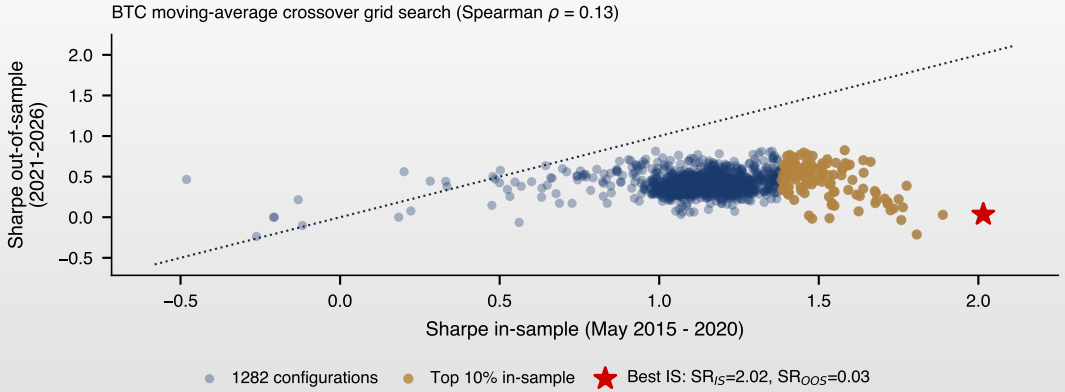
MFM_ch13_backtest_overfitting ▶ Interactive

# In-Sample (IS) vs. Out-of-Sample (OOS): Bitcoin Grid Search (1/2)

- Moving-average (MA) crossover on Bitcoin
  - long when the fast MA of the log price exceeds the slow MA by more than a band
- 1,284 configurations (fast, slow, band)
  - in sample (IS) 16 May 2015 – 2020, out of sample (OOS) 2021–2026
  - the tested family, fixed in sample only: the 1,282 rules with non-zero in-sample variance
  - 3 of them stay in cash throughout the OOS period: OOS SR undefined, counted as 0
  - the decision at the close of day $t$ is traded at the close of $t+1$
- Next chart: IS Sharpe ratio on the horizontal axis, OOS Sharpe ratio on the vertical axis
  - brown points: the top 10% in sample; red star: the selected rule; dotted line: equal SRs
- Selected rule (11/20 days): $\mathrm{SR}_{IS}$ **2.02** $\rightarrow$ $\mathrm{SR}_{OOS}$ **0.03** (annualised with $\sqrt{365}$)
- Spearman rank correlation between IS and OOS SR: $\rho = 0.13$
- Median OOS SR 0.50 for the top 10% in sample vs. 0.41 for all rules
- The result is consistent with overfitting of the selected rule
- A positive family SR does not show incremental predictive information
  - skill against buy-and-hold needs a separate test (SPA, in Further Topics)
- What the result supports: comparison of procedures (in-sample choice, judged out of sample)

MFM_ch13_backtest_overfitting

# In-Sample (IS) vs. Out-of-Sample (OOS): Bitcoin Grid Search (2/2)




MFM_ch13_backtest_overfitting ▶ Interactive

# Sharpe Ratio Inference: Standard Error and PSR (1/2)

- The question
  - how precise is an estimated Sharpe ratio?
  - how do we compare it with a benchmark?
- Per-period Sharpe ratio of daily log returns, risk-free rate 0: $\widehat{SR} = \bar{r}/\hat{\sigma}$
- Shape of the distribution, with $\mu$, $\sigma$ the mean and SD of $r$
  - skewness $\gamma_3 = \mathbb{E}[(r-\mu)^3]/\sigma^3$
  - ordinary kurtosis $\gamma_4 = \mathbb{E}[(r-\mu)^4]/\sigma^4$ (3 for the Normal distribution)
- Sampling variance of $\widehat{SR}$, i.i.d. returns, delta method (Lo (2002); Opdyke (2007)):

$$\mathrm{Var}(\widehat{SR}) \approx \frac{1}{T}\left(1 - \gamma_3\, SR + \frac{\gamma_4 - 1}{4}\, SR^2\right)$$

  - $T$: number of returns; $SR$: true per-period Sharpe ratio
  - negative skewness and fat tails widen the uncertainty; Normal distribution ($\gamma_3 = 0$, $\gamma_4 = 3$): $(1 + \frac{1}{2}SR^2)/T$
  - the formula requires a finite fourth moment, which Chapter 1 questions

# Sharpe Ratio Inference: Standard Error and PSR (2/2)

## Definition 5 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

- Called by its authors the probability that the true SR exceeds a benchmark $SR^*$
  - formally $\Phi$ of a $z$-statistic, with the variance above in the denominator:

$$\widehat{PSR}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{SR}^2}}\right)$$

- $SR^*$: the benchmark, e.g. 0 or the expected maximum $SR_0$
  - $\hat{\gamma}_3$, $\hat{\gamma}_4$: sample skewness and kurtosis
- Reading: $\widehat{PSR} = 1 - p$, with $p$ the one-sided p-value of $H_0 : SR \leq SR^*$
- $\widehat{PSR} > 0.95$ means rejecting $H_0$ at 5%, not a 95% probability for the true SR
  - under i.i.d. returns and the delta-method approximation

# Deflated Sharpe Ratio: the Best Bitcoin Rule

- **Deflated Sharpe Ratio** (DSR, Bailey & López de Prado (2014)): $DSR = \widehat{PSR}(SR_{\mathbf{0}})$
  - the PSR with the benchmark set to $SR_{\mathbf{0}}$, the expected maximum of the False Strategy Theorem
  - $SR_{\mathbf{0}}$ computed with $\mathrm{Var}[\widehat{SR}_n]$ of the trials
  - corrects for selection ($N$ trials), non-Normality and sample length; always report $N$
  - $DSR \leq 0.95$: insufficient evidence at the chosen threshold, not proof that the strategy has no skill
- Worked example, best in-sample rule: $T = 2{,}057$ days
  - $\widehat{SR} = 0.1055$ a day (2.02 annualised), $\hat{\gamma}_{\mathbf{3}} = 0.16$, $\hat{\gamma}_{\mathbf{4}} = 12.23$
  - $\hat{\gamma}_{\mathbf{4}}$ and the DSR require a finite fourth moment, which Chapter 1 questions
  - denominator $\sqrt{1 - 0.16 \cdot 0.1055 + \frac{\mathbf{12.23}-\mathbf{1}}{\mathbf{4}} \cdot 0.1055^{\mathbf{2}}} = 1.007$; $N = 1{,}282$ trials
  - $z = (\widehat{SR} - SR_{\mathbf{0}})\sqrt{T-1}/1.007$, the argument of $\Phi$; $DSR = \Phi(z)$; $\mathrm{SD} = \sqrt{\mathrm{Var}[\widehat{SR}_n]}$

| $\mathrm{Var}[\widehat{SR}_n]$ from | SD | $SR_{\mathbf{0}}$ (per day) | $z$ / DSR |
|---|---|---|---|
| dispersion of the 1,282 trial SRs | 0.0109 | 0.036 | 3.11 / 0.999 |
| independent trials, $1/T$ | 0.0220 | 0.073 | 1.45 / 0.926 |

- Row 1: the observed dispersion of the 1,282 SRs
  - it includes the correlation between rules
- Row 2: rules treated as independent, $\mathrm{Var} = 1/T$, the conservative variant
- The verdict depends on this choice: the out-of-sample SR of 0.03 sides with the conservative value

# Inference after Variable Selection: Post-Double Selection (1/2)

- The question of this section
  - how do we get a valid confidence interval for the effect of one variable after ML has chosen the controls?
- The LASSO path answers "which feature enters first", not "which feature matters"
  - OLS $t$-statistics after LASSO ignore the selection step
- A control correlated with the regressor of interest $d_t$ but only weakly with $y_t$ gets dropped
  - its effect moves into the coefficient of $d_t$ (omitted-variable bias)
- Partially linear model: the effect of $d_t$ is linear
  - the controls enter through unknown functions

$$y_t = \theta\, d_t + g(\mathsf{x}_t) + \varepsilon_t, \qquad d_t = m(\mathsf{x}_t) + v_t$$

  - $y_t$: outcome; $d_t$: regressor of interest; $\theta$: its effect, the parameter to estimate
  - $\mathsf{x}_t$: the $p$ controls; $g$: their effect on $y_t$; $m$: how they predict $d_t$; $\varepsilon_t$, $v_t$: errors
  - approximately sparse: a few controls explain most of $g$ and $m$

# Inference after Variable Selection: Post-Double Selection (2/2)

- Post-double selection (Belloni, Chernozhukov & Hansen (2014)):
  1. LASSO of $y_t$ on $x_t \Rightarrow$ selected set $\hat{S}_y$
  2. LASSO of $d_t$ on $x_t \Rightarrow$ selected set $\hat{S}_d$
  3. OLS of $y_t$ on $d_t$ and $x_{\hat{S}_y \cup \hat{S}_d}$, robust standard error
- $\hat{S}_y$, $\hat{S}_d$: indices of the controls kept by each LASSO
  - $x_{\hat{S}_y \cup \hat{S}_d}$: the controls kept by either
- A control is missed only if it matters for *neither* equation
  - so the omitted bias is second order
- Valid under independent sampling and sparsity
  - with HAC errors an approximation for time series
  - penalty and conditions in the Appendix

# Double/Debiased Machine Learning (DML): Orthogonal Scores and Cross-Fitting

- Chernozhukov et al. (2018): residualise both variables with *any* learner (LASSO, RF, boosting)
  - then regress residual on residual

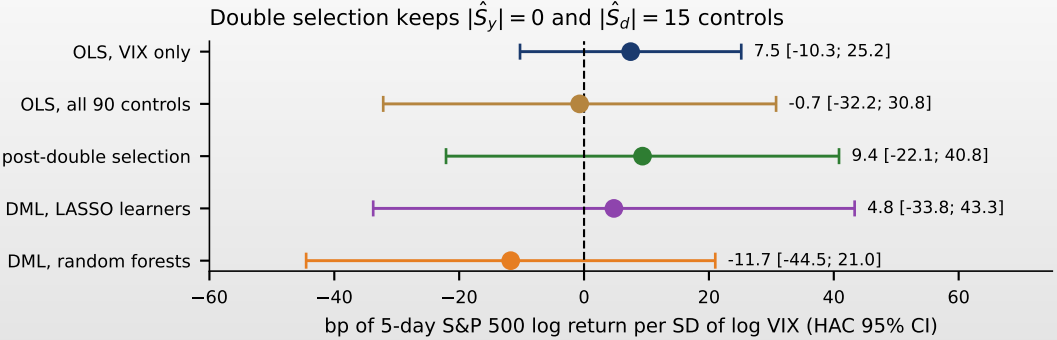
$$\tilde{y}_t = y_t - \hat{\ell}(\mathrm{x}_t), \quad \tilde{v}_t = d_t - \hat{m}(\mathrm{x}_t), \qquad \hat{\theta} = \frac{\sum_t \tilde{v}_t \tilde{y}_t}{\sum_t \tilde{v}_t^2}$$

  - $\hat{\ell}(\mathrm{x}) = \hat{\mathbb{E}}[y \mid \mathrm{x}]$: ML prediction of $y_t$ from the controls; $\hat{m}(\mathrm{x})$: ML prediction of $d_t$
  - $\tilde{y}_t$, $\tilde{v}_t$: the parts of $y_t$ and $d_t$ the controls do not explain; $\hat{\theta}$: OLS slope of $\tilde{y}_t$ on $\tilde{v}_t$
- Neyman-orthogonal score $\psi = (\tilde{y} - \theta\tilde{v})\,\tilde{v}$
  - its mean is zero at the true $\theta$
  - first-order errors in $\hat{\ell}$ and $\hat{m}$ do not bias $\hat{\theta}$ (rate conditions in the Appendix)
- Cross-fitting: fit $\hat{\ell}$, $\hat{m}$ on $K-1$ folds, residualise the held-out fold, repeat
  - removes the own-observation overfitting bias
- The published theory assumes independent sampling
  - with overlapping 5-day labels we use contiguous folds purged by 5 days around the test fold and HAC standard errors
  - an adaptation to serial dependence outside the published theory

# Worked Example: Does the VIX Predict S&P 500 Returns? (1/3)

- The question
  - does the VIX level predict the S&P 500 return over the next 5 days, given a large set of controls?
- $y_t$: 5-day forward log return, in basis points (bp)
  - $d_t$: log VIX, standardised (1 SD = 0.356)
- 90 controls: 12 technical features, their squares and pairwise products
  - $n = 6{,}269$ days; HAC errors with 4 lags
- The single LASSO drops the VIX and every control
  - "no effect", with no standard error
- Double selection keeps $|\hat{S}_y| = 0$ controls in the $y_t$ equation and $|\hat{S}_d| = 15$ in the $d_t$ equation
  - $|\cdot|$: number of elements
- Next chart: one estimator per row
  - the dot is the estimate of $\theta$ (bp per week per SD of log VIX)
  - the bar its 95% HAC CI, the dashed line no effect

# Worked Example: Does the VIX Predict S&P 500 Returns? (2/3)




MFM_ch13_inference ▶ Interactive

# Worked Example: Does the VIX Predict S&P 500 Returns? (3/3)

- Double selection: 95% CI for $\theta$ $[-22.1;\ 40.8]$
  - in bp per week per SD of log VIX
- OLS with log VIX only: 7.5 $[-10.3;\ 25.2]$
  - OLS with all 90 controls, double selection and DML give $-0.7$, 9.4, 4.8 (LASSO) and $-11.7$ (random forests)
  - all with intervals about twice as wide
- Controls that predict $d_t$ absorb part of the variation of log VIX
  - less own variation in $d_t$ means a wider interval
  - the price of protection against omitted-variable bias
- The sign change under DML-RF is noise
  - every point estimate lies inside the double-selection and DML intervals
- Honest intervals contain zero and large effects of both signs
  - the data cannot decide
- What the result supports: association
  - a predictive regression: even a non-zero $\theta$ would not show that the VIX causes returns

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# The Virtue of Complexity, and a Critical Reading [deeper]

- Kelly, Malamud & Zhou (2024): ridge regressions on $P$ random Fourier features of a few predictors
  - rolling windows as short as $T = 12$ months, $P$ in the thousands
  - random Fourier features: sin and cos of random linear combinations of the predictors
  - theory: with $P > T$ the minimum-norm solution (the exact fit with the smallest $\|\boldsymbol{\beta}\|_2$) shrinks implicitly ("double descent")
  - $R^2_{OOS}$ may be negative while the Sharpe ratio of market timing **rises** with complexity; complexity is not automatically overfitting
- Nagel (2025): for $P \gg T$ the forecast is a similarity-weighted average of the $T$ training returns
  - in short windows similarity is mostly recency, so the method becomes a **volatility-timed momentum** strategy, which loses on artificial data with reversals
- Avramov, Cheng & Metzker (2023): deep-learning signal profits come from hard-to-arbitrage stocks
  - excluding microcaps, distressed firms or high-volatility episodes attenuates them, and costs erode them further
- Chen, Pelger & Zhu (2024): impose economic structure instead
  - the no-arbitrage moment conditions become the loss of a deep network (next slides)
- Survey of the field: Kelly & Xiu (2023); textbook: Dixon, Halperin & Bilokon (2020)

## Case Study: Deep Learning in Asset Pricing, Question and Method [deeper]

- Chen, Pelger & Zhu (2024): estimate the stochastic discount factor (SDF) $M_{t+1}$ that prices **all individual US stocks**
  - conditioning on 46 firm characteristics and on the history of 178 macroeconomic series
- No arbitrage: the SDF prices every excess return (Section I.A, eq. 1)

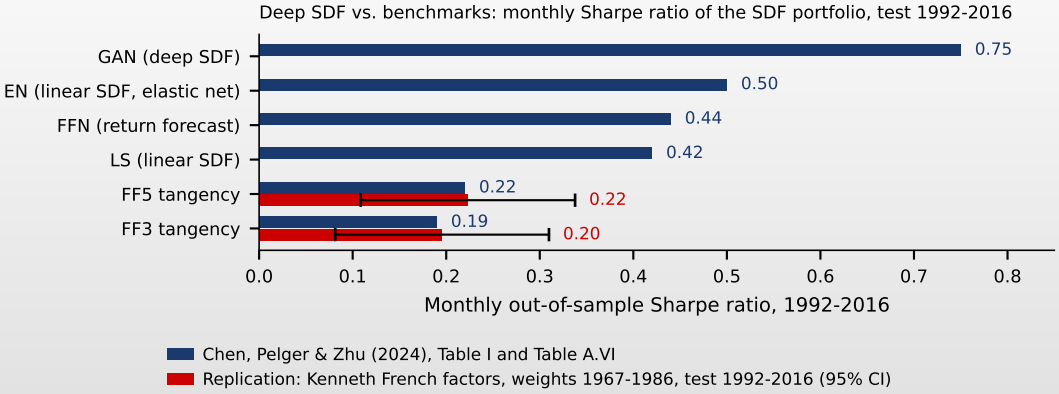
$$E_t[M_{t+1}R^e_{t+1,i}] = 0, \qquad M_{t+1} = 1 - \omega_t^\top R^e_{t+1}$$

  - $R^e_{t+1,i}$: excess return of stock $i$ over the Treasury bill from month $t$ to $t+1$; $R^e_{t+1}$: the vector of all stocks
  - $E_t$: expectation given the information at $t$; $\omega_t$: portfolio weights, a function of characteristics and macro states
- $F_{t+1} = \omega_t^\top R^e_{t+1}$, the SDF portfolio, is the conditional tangency portfolio
  - tangency portfolio: highest Sharpe ratio; and

$$E_t[R^e_{t+1,i}] = \beta_{t,i}\, E_t[F_{t+1}]$$

  - $\beta_{t,i}$: loading of stock $i$ on $F$; a stock's expected excess return is its loading times the premium of $F$ (derivation in Further Topics)

# Deep Learning in Asset Pricing: Out-of-Sample Sharpe Ratios (1/2) [deeper]




MFM_ch13_sdf_replication ▶ Interactive

# Deep Learning in Asset Pricing: Out-of-Sample Sharpe Ratios (2/2) [deeper]

- Blue: the paper's figures (published version), monthly out-of-sample SR 1992–2016
  - GAN 0.75, EN 0.50, FFN 0.44, LS 0.42 (Table I)
  - FF5 0.22, FF3 0.19 (Table A.VI)
  - the regularised no-arbitrage model already beats the forecasting network
- Red: FF3 and FF5 tangency portfolios $w \propto \hat{\Sigma}^{-1}\hat{\mu}$ on the Kenneth French factors
  - weights fixed on 1967–1986
  - FF5 0.22 [0.11; 0.34], FF3 0.20 [0.08; 0.31] (SE 0.058, Lo (2002))
  - the SE requires a finite fourth moment, which Chapter 1 questions
  - $\hat{\mu}$, $\hat{\Sigma}$: sample mean vector and covariance matrix of the factor returns
- What the result supports: comparison of procedures (out-of-sample SRs of the models)
  - after 2016 the same fixed weights give a monthly SR of 0.06 (FF5) and 0.03 (FF3): Further Topics

MFM_ch13_sdf_replication

# An Open Question: Discovery or Data Mining at Scale?

- Chen, Lopez-Lira & Zimmermann (2022): mining 29,000 accounting ratios for $t > 2$
  - gives cross-sectional predictability similar to peer-reviewed predictors
  - for both, about half remains after the original samples
- Novy-Marx & Velikov (2025): an LLM (large language model) pipeline mines over 30,000 signals
  - keeps 96 that pass a fixed protocol
  - writes hundreds of complete papers, with invented motivations and some imagined citations
  - HARKing (hypothesising after the results are known) at scale
- Open: when a machine proposes features, model and story
  - what is the number of trials?
  - what separates a discovery from the best of many draws?
- The discovery loop
  - literature → competing hypotheses → data and code → robustness fixed in advance
  - → referee-style critique → report every test
  - reusable prompts in the ▸ Appendix
- What the human checks
  - every reference resolves (DOI, arXiv)
  - every trial is logged, so $M$ in the correction counts them all
  - correlated tests get a common yardstick (a placebo or a joint bootstrap of the scores)
  - transformations live inside the training fold

# Mini-Case: 102 Tests, One Old Effect Passes the Placebo (1/2)

- A typical assistant proposal: "add more indicators"
  - "momentum, moving-average distances, volatility ratios, RSI and VIX changes over several windows"
- 34 features $\times$ 3 horizons (1, 5, 20 days) = 102 predictive regressions
  - on 2001–2015, Newey–West $t$; only targets whose window ends by 31 December 2015 (1, 5 and 20 days dropped at the boundary)
  - regression $j$: $r_{t,t+h} = \alpha_j + \beta_j x_{j,t} + e_t$, with $x_{j,t}$ feature $j$ and $r_{t,t+h}$ the next $h$-day return; $H_{0,j} : \beta_j = 0$
- Corrections for $M = 102$ tests, $p_{(1)} \leq \cdots \leq p_{(M)}$ the ordered $p$-values
  - Holm: reject while $p_{(j)} \leq 0.05/(M - j + 1)$; controls the FWER
  - Benjamini–Hochberg (BH, Benjamini & Hochberg (1995)): reject the $j$ smallest, for the largest $j$ with $p_{(j)} \leq j\, q/M$
  - controls the false discovery rate (FDR) at $q$, the expected share of false rejections among all rejections
  - shift placebo: the 95% quantile of $\max_j |t_j|$ when the target is shifted circularly
  - the shift keeps the correlation between the features but also changes their link with conditional volatility: a heuristic yardstick, not an exact test

# Mini-Case: 102 Tests, One Old Effect Passes the Placebo (2/2)

- $|t| > 1.96$: **22** of 102, against 5.1 expected under the global null at 5%
- Holm (FWER 5%): **0**; BH (FDR 10%): **13**; $|t| > 3$ (Harvey, Liu & Zhu (2016)): **4**
- Shift placebo: 95% quantile of $\max_j |t_j| = 3.32$
  - 500 circular shifts of the target
  - one predictor exceeds it, the 1-day return on the next day's return, $t = -3.38$: short-term reversal, a known effect
- Holdout 2016–2026: 3 of the 22 keep their sign with $|t| > 1.96$
  - nominal replications, all three reversal variants; the VIX-based "finds" fail
  - with Holm at 5% over the 22 checks none passes (smallest $p$ 0.019, threshold $0.05/22 = 0.0023$)
- BH needs independence or positive dependence of the true-null p-values
  - correlated predictors do not ensure it
- What the result supports: association
  - short-term reversal is a predictive regularity, not an identified mechanism

MFM_ch13_ai_discovery

# Your Turn: a Project Seed

- **Questions**
  - can an AI-assisted feature search find a predictor of index returns (S&P 500, BET – the Bucharest Stock Exchange benchmark index, Bitcoin)?
  - does it survive a pre-registered correction and a holdout?
  - does it go beyond short-term reversal?
- **Data**: S&P 500 and Bitcoin are in the course data; BET is not
  - a team that chooses BET names the source (e.g. the Bucharest Stock Exchange) and freezes the sample
  - it logs the download date, $n$ and the first and last close
- **Deliverables**
  - a feature list fixed before the data are touched
  - all tests with Holm, BH and the max-$|t|$ null
  - a separate holdout
  - DSR and PBO for any strategy built on the survivors
- **Log** (templates)
  - each prompt that shaped a decision, in `AI_USE.md`
- Each invented reference, leak or undercounted trial in `AI_ERRORS.md`, with the test that caught it

# Key Takeaways

## Four empirical lessons

- An ADF rejection is not stationarity
  - truncated FFD of the S&P 500 log price stays $I(1)$
  - estimate memory directly with local Whittle
- Shuffled $K$-fold "finds" skill in pure noise
- On the S&P 500 no model beats "always up" (PT, DM, GW)
- The best of many backtests must be deflated or bootstrapped
  - on the Bitcoin grid the best in-sample rule keeps nothing out of sample

## Next

- Chapter 14
  - deep learning and time-series foundation models
- Chapter 15
  - large language models and sentiment

## A Checklist Before Any Claim

- ☐ Point-in-time data; transformations fitted on training folds only
- ☐ Volatility-scaled labels with recorded spans
- ☐ Purged $K$-fold or CPCV with embargo for selection; walk-forward for the final test
- ☐ A naive baseline; PT, DM / CW / GW and block-bootstrap intervals
- ☐ Costs included and every trial logged
- ☐ DSR, SPA and PBO for the selected strategy
- ☐ "Selected" or "important" is not "significant": double selection or DML for inference

## Check Your Understanding

1. Why does the logit with 57.7% accuracy show no skill when 58.1% of the days are "up"?
2. What does a correlation of 0.98 between the FFD series ($d = 0.25$) and the log price say about an ADF rejection?
3. Why does shuffled $K$-fold report 69.4% on a target with nothing to predict, while purged $K$-fold with embargo reports 49.6%?
4. How does $SR_0$, the expected maximum of $N$ zero-skill strategies, change when $N$ rises from 10 to 1,000?
5. Why does the best Bitcoin rule, with an in-sample SR of 2.02, get a DSR of 0.999 or 0.926 depending on the variance used?
6. Why do the DeLong interval for the AUC, [0.466; 0.494], and the block-bootstrap interval, [0.456; 0.505], lead to different conclusions?
7. Which question about the VIX does double selection answer that the LASSO path does not?

# The Practical Decision

- **What I choose**
  - for the 5-day direction of the S&P 500: buy-and-hold, not the GBM strategy
  - in the 2010–2026 walk-forward, the Sharpe ratio is 0.66 for buy-and-hold and 0.33 for GBM
  - for any model: purged CV with embargo for selection, walk-forward for the final evaluation
- **Under which conditions**
  - the directional signal stays weak: no model beats the "always up" benchmark of 58.1%
  - each change of position costs 5 bp
  - every trial is logged and the number of trials $N$ is reported
- **What result would make me drop it**
  - a model that beats buy-and-hold out of sample, after costs
  - with a PT statistic above 1.645
  - with a block-bootstrap AUC interval entirely above 0.5
  - with a DSR above 0.95 under the conservative variance $1/T$ (the best Bitcoin rule has 0.926)

# Evidence, and the Economic Value of a Small $R^2_{OOS}$ (1/2)

- Campbell & Thompson (2008): what a small $R^2$ is worth to an investor
  - returns have a predictable and an unpredictable part

$$r_{t+1} = \mu + x_t + \varepsilon_{t+1}, \qquad R^2 = \frac{\mathrm{Var}(x_t)}{\mathrm{Var}(r_{t+1})}$$

  - $\mu$: mean return; $x_t$: predictable component, known at $t$, mean zero; $\varepsilon_{t+1}$: unpredictable shock
  - $R^2$: share of the return variance explained by the predictor
- An unconstrained mean-variance investor who knows $x_t$ and re-optimises every period
  - no costs, no estimation error; $SR_t$: the conditional Sharpe ratio at $t$

$$\mathbb{E}[SR_t^2] = \frac{SR_{BH}^2 + R^2}{1 - R^2}$$

  - $\mathbb{E}[SR_t^2]$ is not, in general, the square of the unconditional Sharpe ratio of the dynamic strategy
  - the optimal mean-variance utility at $t$ is $SR_t^2/(2\gamma)$, $\gamma$ the risk aversion, so the gain reads as mean utility
  - $SR_{BH}$: SR of buy-and-hold, which ignores $x_t$; $R^2 = 0$ gives $\mathbb{E}[SR_t^2] = SR_{BH}^2$

MFM_ch13_inference

# Evidence, and the Economic Value of a Small $R^2_{OOS}$ (2/2)

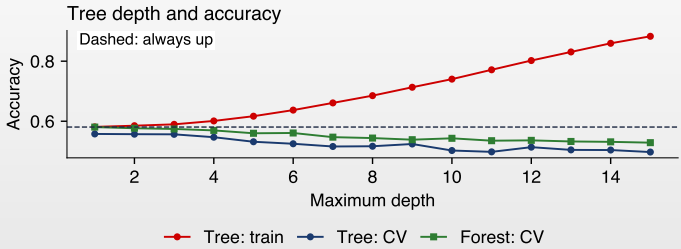
- S&P 500, monthly log returns 2001–2026
  - $SR_{BH} = 0.128$ a month (0.44 annualised with $\sqrt{12}$)
  - $R^2 = 0.5\%$ a month $\Rightarrow \sqrt{\mathbb{E}[SR_t^2]} = 0.51$ annualised ($+15\%$ over $SR_{BH}$)
  - $R^2 = 1\% \Rightarrow 0.57$ (about $+27\%$)
- For a **single index** the effective sample is far smaller than for thousands of stocks
  - the situation of the S&P 500 case study

MFM_ch13_inference

## Where the Campbell–Thompson Formula Comes From

- Same model: $r_{t+1} = \mu + x_t + \varepsilon_{t+1}$
  - $\mathbb{E}[x_t] = 0$, $\varepsilon_{t+1}$ independent of $x_t$
  - write $\sigma_r^2 = \mathrm{Var}(r_{t+1})$
- Step 1, the risk the informed investor still bears
  - $\mathrm{Var}(r_{t+1}) = \mathrm{Var}(x_t) + \sigma_\varepsilon^2$ and $R^2 = \mathrm{Var}(x_t)/\sigma_r^2$, so
$$\sigma_\varepsilon^2 = (1 - R^2)\,\sigma_r^2$$
- Step 2, the investor's conditional Sharpe ratio at $t$
  - mean $\mu + x_t$, variance $\sigma_\varepsilon^2$, squared SR $(\mu + x_t)^2/\sigma_\varepsilon^2$
- Step 3, average over $t$: $\mathbb{E}[(\mu + x_t)^2] = \mu^2 + \mathrm{Var}(x_t) = \mu^2 + R^2\sigma_r^2$
- Step 4, divide and use $SR_{BH} = \mu/\sigma_r$:
$$\mathbb{E}[SR_t^2] = \frac{\mu^2 + R^2\sigma_r^2}{(1 - R^2)\,\sigma_r^2} = \frac{SR_{BH}^2 + R^2}{1 - R^2}$$
- So the formula gives the average squared *conditional* SR, not the unconditional SR of the strategy
  - for an investor who re-optimises each month, in monthly units
- Numbers, $R^2 = 0.005$
  - $\mathbb{E}[SR_t^2] = (0.128^2 + 0.005)/0.995 = 0.02149$, $\sqrt{\mathbb{E}[SR_t^2]} = 0.1466$ a month
  - $\times\sqrt{12} = 0.51$ a year; $0.1466/0.128 = 1.15$

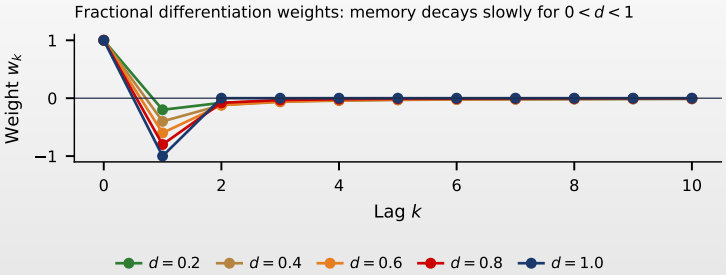
# Bias–Variance in Practice: Tree Depth




- S&P 500, 5-day direction: training accuracy rises with depth to $\approx 88\%$ at depth 15
  - purged-CV accuracy **falls** towards 50%
  - purged CV drops from training the observations whose labels overlap the test set (Section 5)
- Horizontal axis: maximum tree depth; vertical axis: accuracy
  - red training, blue CV of one tree, green CV of the forest
- Random forests degrade more slowly, but never beat the "always up" baseline (dashed line)

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# Fractional Differentiation Weights




- Horizontal axis: lag $k$; vertical axis: weight $w_k$, for five values of $d$
- For $d = 1$ only the first lag matters
- For small $d$ the weights of distant lags remain non-zero
- All weights after $w_0$ are negative
  - $\tilde{X}_t$ compares today's level with a weighted history

MFM_ch13_ffd

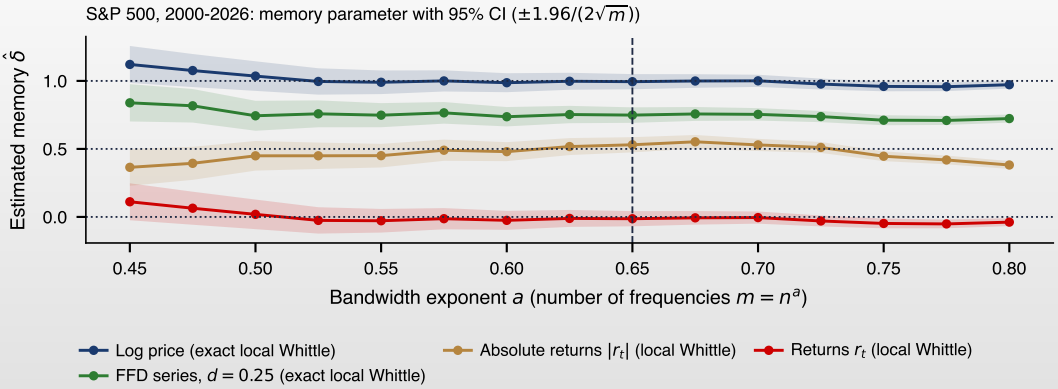
# Memory Estimated Directly: Local Whittle (1/3)

- Idea: a long-memory series has most of its variance in slow cycles
  - the slope of its spectrum near frequency 0 reveals $\delta$

$$f(\lambda) \sim G\,\lambda^{-2\delta} \quad \text{as } \lambda \to 0$$

  - $f(\lambda)$: spectral density, the variance contributed by cycles of frequency $\lambda$ (period $2\pi/\lambda$ days)
  - $G > 0$: scale constant; $\sim$: the ratio of the two sides tends to 1
  - $\delta > 0$: $f$ explodes near 0 (long memory); $\delta = 0$: flat (short memory); $\delta < 0$: $f$ vanishes near 0
- Local Whittle (LW, Robinson (1995)): fits this law only on the first $m = n^a$ Fourier frequencies
  - $\lambda_j = 2\pi j/n$, $j = 1, \ldots, m$
  - $n$: sample size; $a \in (0, 1)$: bandwidth exponent; a larger $a$ uses more frequencies: less variance, more bias from short-run dynamics
- Exact local Whittle (ELW, Shimotsu & Phillips (2005)): also valid for non-stationary $\delta \geq \frac{1}{2}$
  - both estimators in the Appendix
- 95% confidence interval (CI): $\hat{\delta} \pm 1.96/(2\sqrt{m})$
  - the standard error $1/(2\sqrt{m})$ depends only on $m$

# Memory Estimated Directly: Local Whittle (2/3)



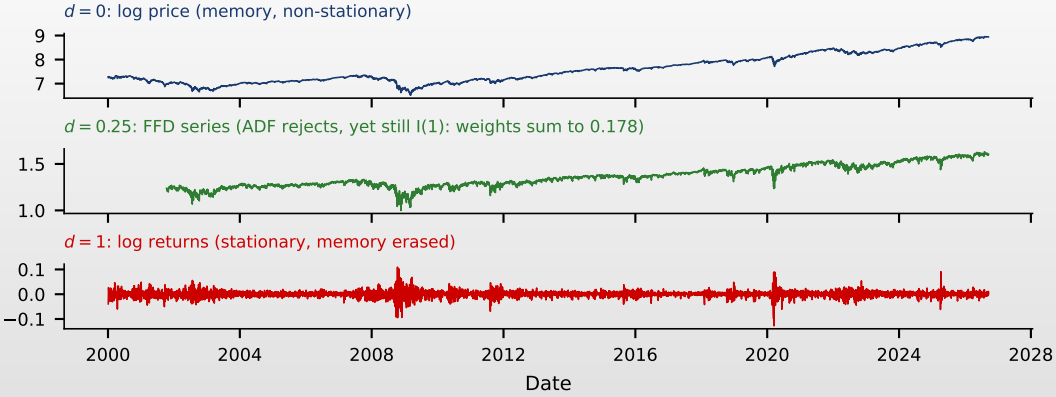

MFM_ch13_ffd

## Memory Estimated Directly: Local Whittle (3/3)

- Chart: the exponent $a$ (with $m = n^a$) on the horizontal axis, $\hat{\delta}$ on the vertical axis
  - with its 95% CI band
  - the vertical line marks $a = 0.65$, the value used below (S&P 500, 2000–2026)
- Log price (ELW): $\hat{\delta} = 0.99$, CI [0.94; 1.05]
- FFD series (ELW): $\hat{\delta} = 0.75$, CI [0.69; 0.81]
- Returns: $\hat{\delta} = -0.01$, CI [−0.07; 0.04]
- Absolute returns $|r_t|$: $\hat{\delta} = 0.53$, CI [0.47; 0.59]
- The FFD series stays near $1 - d$ over the frequency band used, although its integration order is still 1
- The long memory is in **volatility** (Chapters 5 and 9)
  - so fractional filters make sense for $|r_t|$, not for the price

MFM_ch13_ffd

# Prices, FFD Series and Returns (1/2)



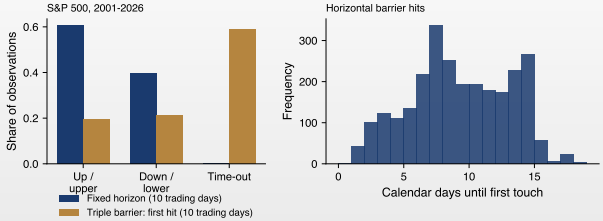

MFM_ch13_ffd

## Prices, FFD Series and Returns (2/2)

- The top panel shows the S&P 500 log price, an $I(1)$ series
- The middle panel shows the FFD series with $d = 0.25$: the price trend is still visible
  - because the series is $0.178 \log P_t$ plus a stationary part
- The bottom panel shows log returns, with no trend
  - only volatility clustering is visible

MFM_ch13_ffd

# Label Distribution: Fixed Horizon vs. Triple Barrier




- S&P 500, 2001–2026, $h = 10$, barriers at $\pm 1\,\sigma\sqrt{h}$
  - 19.6% up, 21.3% down, 59.0% time-out
  - labels after the sign rule at the vertical barrier: 59.1% $+1$ and 40.9% $-1$ (fixed horizon: 60.5% up)
- The chart counts **exit types**: time-out = exit at the vertical barrier
  - under the sign rule these 59.0% still get $\pm 1$ from the sign of their return
  - under the "0" convention they form a third class
- Right panel: histogram of calendar days until the first horizontal barrier is touched
  - typically after 1–2 weeks

MFM_ch13_triple_barrier

# Overlapping Labels and Sample Uniqueness (1/2)

- Label $i$ spans $[t_{0,i}, t_{1,i}]$
  - consecutive labels share returns $\Rightarrow$ samples are **not i.i.d.**
- Concurrency: the number of labels whose span contains day $t$

$$c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$$

- Average uniqueness of label $i$: the average share of each day of its span that belongs to it alone

$$\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$$

  - $\bar{u}_i = 1$: no shared return; $\bar{u}_i = 1/5$: the harmonic mean of $c_t$ over the span of label $i$ is five

# Overlapping Labels and Sample Uniqueness (2/2)

## Consequences

- Effective sample size $\approx \sum_i \bar{u}_i \ll n$
  - $n$: the number of labels
  - an overlap heuristic, not a variance-based measure
- Bagging draws near-duplicates
  - CV scores too optimistic

## Remedies

- Weight samples by $\bar{u}_i$ (and by $|r|$)
- Sequential bootstrap
  - in RF, the fraction of observations drawn for each tree $\approx \bar{u}$, the mean uniqueness

## Transformations on Training Data Only: the FFD Ablation

- The "minimum $d$" was chosen on 2000–2026 and then used inside purged CV
  - the test folds helped choose a feature
  - the same ADF rule on data up to 2009 gives $d = 0.15$; on the full sample $d = 0.25$
- Case study without the FFD log-price (purged 5-fold, mean over folds)
  - accuracy: logit 57.5%, RF 56.4%, GBM 55.6%, MLP 56.5%, all below 58.1%
  - AUC: logit 0.483, RF 0.520, GBM 0.514, MLP 0.526
- The negative conclusion does not depend on this feature
- Correct protocol
  - estimate $d$ (local Whittle, with its CI) inside each training fold only
  - prefer features with an economic meaning: distances to moving averages, fractional filters of $|r_t|$

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# Overlapping Labels and the Newey–West Long-Run Variance

- The significance tests of the case study (Pesaran–Timmermann) average a daily series $z_t$
  - $z_t$: a hit, a loss differential $d_t$, a regression score
  - with 5-day labels, neighbouring $z_t$ share returns, so $\mathrm{Var}(\bar{z}) \neq \mathrm{Var}(z_t)/n$
- Newey–West (Newey & West (1987)) heteroskedasticity and autocorrelation consistent (HAC) long-run variance
  - with Bartlett weights:

$$\hat{\Omega}_{NW} = \hat{\Gamma}_0 + 2\sum_{j=1}^{q}\Big(1 - \frac{j}{q+1}\Big)\hat{\Gamma}_j, \qquad \hat{\Gamma}_j = \frac{1}{n}\sum_{t=j+1}^{n}(z_t - \bar{z})(z_{t-j} - \bar{z})$$

  - $\hat{\Gamma}_j$: sample autocovariance of $z_t$ at lag $j$; $q$: number of lags ("lag 4" in the tables); $\widehat{\mathrm{SE}}(\bar{z}) = \sqrt{\hat{\Omega}_{NW}/n}$
  - lag rule: $h$-day labels share returns up to $h-1$ days apart, so $q \geq h-1 = 4$ is the minimum here
- Toy example: $z_t$ = sum of the next 5 i.i.d. daily returns of variance $\sigma^2$
  - $\Gamma_j = (5-j)\sigma^2$ for $j \leq 4$, 0 beyond
  - i.i.d. formula: $\Gamma_0 = 5\sigma^2$; true long-run variance $\Gamma_0 + 2(4+3+2+1)\sigma^2 = 25\sigma^2$: an i.i.d. $t$ is $\sqrt{5} \approx 2.2$ times too large
  - Bartlett with $q = 4$: $5 + 2(0.8 \cdot 4 + 0.6 \cdot 3 + 0.4 \cdot 2 + 0.2 \cdot 1) = 17\sigma^2$; the weights damp far lags, so $q = h-1$ is a floor, not a generous choice

# Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (1/2)

- Two probability forecasts of $y_t \in \{0, 1\}$
  - model 1 gives $\hat{p}_{1,t}$
  - the benchmark (model 0, e.g. a constant) gives $\hat{p}_{0,t}$
- Loss of a forecast $\hat{p}_t$
  - the log-loss of the toolbox slide, with $f = \hat{p}_t$, or the Brier loss $(y_t - \hat{p}_t)^2$
- Loss differential $d_t = L(y_t, \hat{p}_{1,t}) - L(y_t, \hat{p}_{0,t})$
  - $d_t > 0$: model 1 worse on day $t$
- **Diebold–Mariano** (DM, Diebold & Mariano (1995)): is the mean loss differential zero?

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}_{NW}/n}} \xrightarrow{d} N(0,1) \quad \text{under } H_0 : \mathbb{E}[d_t] = 0$$

  - $\bar{d}$: mean of $d_t$ over the $n$ test days; $\hat{\Omega}_{NW}$: Newey–West long-run variance of $d_t$ (Newey–West formula, with $z_t = d_t$)
  - $DM > 1.96$: model 1 significantly worse at 5%; $DM < -1.96$: significantly better

# Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (2/2)

- **Nested models**: the logit with 14 features nests the constant, $\beta = 0$
  - under $H_0$ the large model estimates zeros with noise, $\mathbb{E}[d_t] > 0$
  - DM therefore rarely favours it
- **Clark–West** (CW, Clark & West (2007)): one-sided HAC $t$-test of $\mathbb{E}[f_t] = 0$ against $\mathbb{E}[f_t] > 0$

$$f_t = (y_t - \hat{p}_{0,t})^2 - \left[(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2\right]$$

  - the Brier-loss gain of model 1, plus the term $(\hat{p}_{0,t} - \hat{p}_{1,t})^2$ that removes the noise cost of estimating zero coefficients; a large positive $t$: the features add information
  - derived for nested linear models estimated by ordinary least squares (OLS); for penalised-logit probabilities an approximation, not a theorem
- **Giacomini–White** (GW, Giacomini & White (2006)): tests the forecasting *method* as deployed
  - with a bounded estimation window
- By contrast, CW asks whether the population model adds information
  - the GW statistic in the Appendix

# Feature Importance: MDI and MDA

- The question
  - which features does the model rely on?
  - which of them help prediction on new data?
- Mean decrease impurity (MDI), trees only, computed in sample:

$$\mathrm{MDI}_j = \frac{1}{B}\sum_{b=\mathbf{1}}^{B}\sum_{v\in T_b:\ j(v)=j}\frac{n_v}{n}\Big[G_v - \frac{n_L}{n_v}\,G_L - \frac{n_R}{n_v}\,G_R\Big]$$

  - the sum of the Gini reductions of the nodes $v$ that split on feature $j$, weighted by the share $n_v/n$ of observations reaching the node
  - favours continuous features with many possible thresholds
- Mean decrease accuracy (MDA), any model, on test data:

$$\mathrm{MDA}_j = S\big(\hat{f};\,\mathcal{D}_{test}\big) - \frac{1}{R}\sum_{r=\mathbf{1}}^{R} S\big(\hat{f};\,\mathcal{D}_{test}^{\pi_r(j)}\big)$$

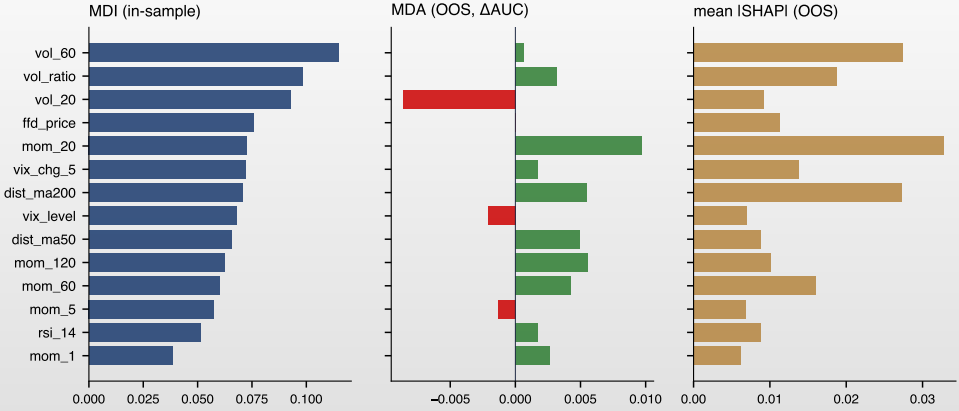
  - $S$: the score (here AUC); $\mathcal{D}_{test}^{\pi_r(j)}$: the test data with column $j$ randomly permuted, repetition $r$
  - $\mathrm{MDA}_j < 0$: permuting raised the score; read it with the spread over the $R$ permutations and the estimation uncertainty
  - a small negative MDA alone does not show that a feature hurts

## Feature Importance: SHAP Values

- SHAP (Lundberg & Lee (2017)): the Shapley value $\phi_j(\mathrm{x})$
  - the average marginal contribution of feature $j$ to the prediction at x
  - averaged over all coalitions of other features (coalition formula: Appendix)
- Additive: $\hat{f}(\mathrm{x}) = \phi_0 + \sum_j \phi_j(\mathrm{x})$, $\phi_0$ the average prediction
- Global importance of feature $j$: $\frac{1}{n}\sum_i |\phi_j(\mathrm{x}_i)|$, computed on test data
- MDI answers "what did the model use?"
  - SHAP attributes the model's prediction across features and does not use the realised outcome
  - MDA measures the change in score on the evaluated set when a feature is permuted
- Correlated features share and dilute importance (substitution)
  - cluster them first (López de Prado (2020))

Shapley Values

# Feature Importance in the Case Study (1/2)




MFM_ch13_feature_importance

# Feature Importance in the Case Study (2/2)

- Fit on the first 70%, evaluated on the last 30% after a purge
  - **RF** gives MDI and MDA, **GBM** gives SHAP
  - so the differences are not purely methodological
- Panels: MDI (in sample), MDA and mean $|\phi_j|$ (out of sample)
  - MDA: out-of-sample AUC drop; red bars are negative
- Labels beyond the LASSO path
  - mom_$h$, $h$-day momentum; dist_ma50, dist_ma200, the distance to the moving average
  - rsi_14, the RSI(14); vix_level, vix_chg_5, the VIX level and its 5-day change
- MDI ranks volatility first
  - MDA and SHAP point to 20-day momentum and the distance from the 200-day moving average
- 20-day volatility: high MDI, **negative** MDA
  - a sign of overfitting, not a proof
- All importances are tiny ($\Delta\text{AUC} < 0.01$)
  - how to read them: Appendix
- Correlated features: MDA and SHAP **extrapolate** (Hooker, Mentch & Zhou (2021))
  - permutation creates off-manifold points

MFM_ch13_feature_importance

## Walk-Forward: Do the Forecasts Beat a Constant?

- Walk-forward 2010–2026, $n = 4{,}198$ days, 60.2% "up"
  - benchmark: the constant forecast = training share of "up"
  - DM, Clark–West and GW use Newey–West with 4 lags
- Logit (nests the constant): log-loss 0.6824 vs. 0.6756
  - DM $t = 2.50$ ($p = 0.012$): the estimated logit is significantly *worse*
  - Clark–West $t = -0.36$ (one-sided $p = 0.64$): no evidence that the features carry information
  - Giacomini–White $= 6.29$ ($\chi^2_2$, $p = 0.043$): the logit is worse at 5% as well; approximate, since the window expands (rolling-window result in the Appendix)
- GBM: log-loss 0.7027, worse than a constant
  - DM $t = 5.81$, GW $= 33.6$ ($p < 0.001$)
  - Pesaran–Timmermann $-2.74$
- Conclusion: no model beats "always up" or a constant forecast

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## Annualising the Sharpe Ratio and Comparing Two Strategies

- Annualising with $\sqrt{A}$ needs zero autocorrelation; otherwise $SR_A = \eta(A)\, SR$ (Lo (2002))
  - $A$: return periods per year (252 trading days; 365 for Bitcoin)
  - $\eta(A) = A/\sqrt{A + 2\sum_{k=1}^{A-1}(A-k)\rho_k}$, $\rho_k$: autocorrelation of returns at lag $k$
  - positive autocorrelations, $\rho_k > 0$, make $\eta(A) < \sqrt{A}$
- Example: monthly returns, $A = 12$, with $\rho_1 = 0.2$ and all other autocorrelations zero
  - $\eta = 12/\sqrt{12 + 2 \cdot 11 \cdot 0.2} = 12/\sqrt{16.4} = 2.96$, against $\sqrt{12} = 3.46$
  - so the annual SR is about 15% lower
- Two strategies on the same days: test $SR_a - SR_b$
  - with the HAC delta method or a studentised block bootstrap (Ledoit & Wolf (2008))
  - not with two separate intervals

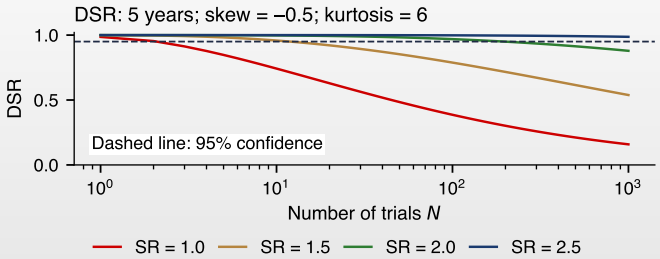
# Correlated Trials in the DSR: Alternative Approximations

## Two approximations with different assumptions

- (a) raw $N$, with $\mathrm{Var}[\widehat{SR}_n]$ = dispersion of the trial SRs: row 1 of the Bitcoin DSR table
- (b) $\mathrm{Var}[\widehat{SR}_n] = 1/T$, with an effective number of independent trials $\hat{N} < N$
  - Seminar 13, exercise B6
- Used together, dispersion *and* $\hat{N}$ can count the same dependence effect twice (toy model below)

- Why (a) works, toy model: $N$ zero-skill trials $\widehat{SR}_n = \sigma(\sqrt{\rho}\,Z + \sqrt{1-\rho}\,e_n)$
  - each of variance $\sigma^2 = 1/T$, independent standard Normal $Z, e_1, \ldots, e_N$, pairwise correlation $\rho$
  - the common shock moves all trials together, so only $\sqrt{1-\rho}\,\sigma\,e_n$ creates selection: $\mathbb{E}[\max_n \widehat{SR}_n] = \sqrt{1-\rho} \times$ the independent-trial value
  - the dispersion across trials is $(1-\rho)\sigma^2$, so (a) captures this scaling
  - the two rows have SD ratio $0.0109/0.0220 = 0.50$, which this toy model reads as $\rho \approx 1 - 0.50^2 \approx 0.75$
- The dispersion also contains true differences between rules
  - with real rules (a) is an approximation
- Recommendation: report the conservative row and either (a) or (b), with $N$
  - call a rule a discovery only if it also passes the conservative row

# Deflated Sharpe Ratio in Practice




- 5 years of daily returns, skewness $-0.5$, kurtosis 6
  - horizontal axis: number of trials $N$ (log scale), vertical axis: DSR, dashed line: the 0.95 threshold
- $N = 1$: no selection, $SR_0 = 0$, so $DSR = \widehat{PSR}(0)$; for an annualised SR of 1.0, DSR $= 0.986$
  - SR $= 1.0$ stays above 0.95 at $N = 2$ (0.954) and falls below from $N = 3$ (0.913)
- SR $= 1.5$ survives $\approx 10$ trials, SR $= 2.0 \approx 200$
  - only SR $= 2.5$ survives 1000

MFM_ch13_backtest_overfitting

# Data-Snooping Tests on the Bitcoin Grid (1/3)

- The question
  - does any of the $K$ rules beat the benchmark, once we account for having tried them all?
- $r_{k,t}$: return of rule $k$ on day $t$; $r_{0,t}$: benchmark return
  - $d_{k,t} = r_{k,t} - r_{0,t}$, with mean $\bar{d}_k$ over the $n$ days
- $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$: no rule beats the benchmark
- Null distribution from a **stationary bootstrap** (Politis & Romano (1994))
  - resample blocks of whole days, of random length
  - $\bar{d}_k^*$: the mean in a bootstrap replication
- The same days for all rules keep the correlation between them
  - no "effective" $N$ needed
- Reality Check (RC, White (2000)): statistic $T^{RC} = \max_k \sqrt{n}\,\bar{d}_k$
  - its p-value is the share of replications with $\max_k \sqrt{n}\,(\bar{d}_k^* - \bar{d}_k) \geq T^{RC}$

## Data-Snooping Tests on the Bitcoin Grid (2/3)

- Superior Predictive Ability (SPA, Hansen (2005)): studentised, $T^{SPA} = \max\left(\max_k \sqrt{n}\,\bar{d}_k/\hat{\omega}_k,\ 0\right)$
  - $\hat{\omega}_k$: the bootstrap SD of $\sqrt{n}\,\bar{d}_k$
  - replications are recentred with $\bar{d}_k\,1\{\sqrt{n}\,\bar{d}_k/\hat{\omega}_k \geq -\sqrt{2\log\log n}\}$: clearly bad rules are not recentred, so adding junk rules does not destroy power
- Romano–Wolf StepM (Romano & Wolf (2005)): names *which* rules beat the benchmark
  - with family-wise error rate (FWER) control, the probability of at least one false rejection
  - step 1: reject $H_{0,k}$ for every $k$ with $\sqrt{n}\,\bar{d}_k/\hat{\omega}_k > \hat{c}_1$, where $\hat{c}_1$ is the $1-\alpha$ quantile of $\max_k \sqrt{n}\,(\bar{d}_k^{*} - \bar{d}_k)/\hat{\omega}_k$
  - step $j$: recompute the quantile on the rules not yet rejected; stop when no further rule is rejected

# Data-Snooping Tests on the Bitcoin Grid (3/3)

- Setup: 1,282 rules, in-sample $n = 2{,}057$ days, mean block 20 days, 2,000 replications

| Benchmark | largest studentised mean | RC $p$ | SPA $p$ | StepM rules kept |
|---|---|---|---|---|
| cash ($r_0 = 0$) | 4.20 | 0.001 | 0.002 | 117 |
| buy-and-hold | 1.36 (the 11/20 rule) | 0.60 | 0.50 | 0 |

- Against cash, some trend rule earns a positive mean: Bitcoin's own trend
  - no rule beats holding Bitcoin, even in sample
  - consistent with the OOS SR of 0.03
- DSR needs $N$ and $\mathrm{Var}[\widehat{SR}]$, hence its two verdicts
  - SPA takes both from the data

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# Probability of Backtest Overfitting (PBO)

## Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix of returns: $T$ periods $\times$ $N$ configurations; split the rows into $S$ contiguous blocks
2. For each of the $\binom{S}{S/2}$ choices, half of the blocks are in sample (IS), the rest out of sample (OOS)
3. Pick the best IS configuration $n^*$
   - rank all $N$ configurations OOS in ascending order
   - set $\bar{\omega} = \text{rank}(n^*)/(N+1) \in (0,1)$
4. Logit $\lambda = \log \frac{\bar{\omega}}{1-\bar{\omega}}$; $PBO = \Pr[\lambda \leq 0]$
   - the share of splits in which the IS winner ranks at or below the OOS median

- $\bar{\omega}$: relative OOS rank of the IS winner, near 1 if it stays the best
  - $\lambda > 0$: above the OOS median
- $PBO \approx 0$: the IS winner reliably stays above the OOS median
  - $PBO \approx 0.5$ or more: selecting the IS winner is no better than picking at random
- PBO evaluates the **research procedure**, not one strategy
  - it is not the probability of losing money
  - like DSR it needs **every** trial logged (Seminar 13, exercise B12)

## From the Pricing Condition to the Beta Representation

- Three steps (written unconditionally; conditionally, replace $E$ by $E_t$)
  1. With $M = 1 - F$: $E[(1 - F)R_i^e] = 0$, so $E[R_i^e] = E[R_i^e F]$ for every stock
  2. $F$ is itself an excess return; apply step 1 to $R_i^e = F$: $E[F] = E[F^2]$
  3. Define $\beta_i = E[R_i^e F]/E[F^2]$; then $E[R_i^e] = E[R_i^e F] = \beta_i E[F^2] = \beta_i E[F]$
- This $\beta_i$ is defined with second moments
  - when step 1 holds it equals the regression beta, since $\mathrm{Cov}(R_i^e, F) = \beta_i E[F^2] - \beta_i E[F]^2 = \beta_i \mathrm{Var}(F)$
  - estimated on data with pricing errors, the two can differ
  - so compute the second-moment form, as the paper's loading network does (Appendix)
- Example: $E[F] = E[F^2] = 0.01$ (1% a month), $E[R_i^e F] = 0.005$
  - $\Rightarrow \beta_i = 0.5$, $E[R_i^e] = 0.5\%$ a month
  - the regression beta gives the same value, $(0.005 - 0.005 \cdot 0.01)/(0.01 - 0.01^2) = 0.5$
- Why the loss changes: any $\omega$ that prices all assets must satisfy step 1 for every $i$
  - step 1 is a first-moment condition
  - this is what the adversarial GMM loss on the next slide enforces

# Case Study: Deep Learning in Asset Pricing, Question and Method (1/2)

- Why: forecasting fits a second moment where signal-to-noise is low
  - forecasting: Gu, Kelly & Xiu (2020), the benchmark feedforward network (FFN)
- The no-arbitrage condition targets a first moment
  - averaging over time removes much of the noise
- Method: adversarial generalized method of moments (GMM, eq. 3, after Hansen & Jagannathan (1997))

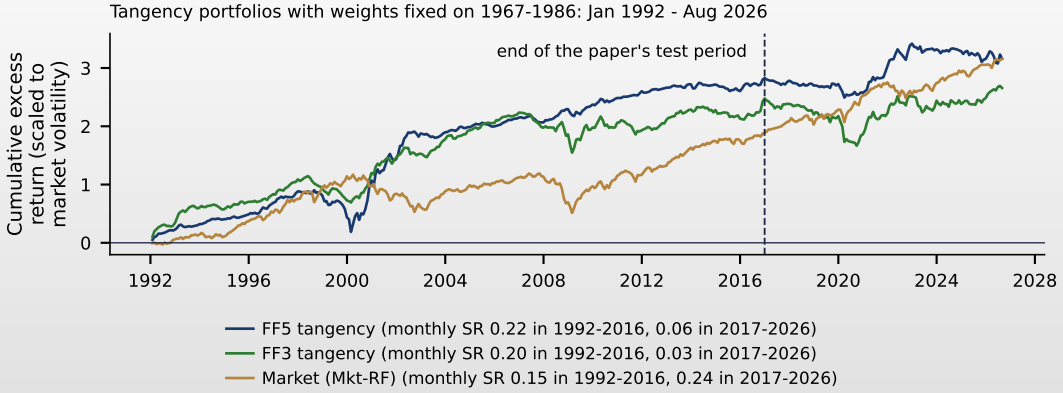
$$\min_{\omega} \max_{g} \frac{1}{N} \sum_{j=1}^{N} \left( E\left[ M_{t+1} R^{e}_{t+1,j} \, g(I_t, I_{t,j}) \right] \right)^2$$

  - $I_t$: macroeconomic information; $I_{t,j}$: characteristics of stock $j$; $N$: number of stocks
  - $g$: instrument function chosen by an adversary to make the pricing errors as large as possible
  - $\omega$ minimises this worst case

# Case Study: Deep Learning in Asset Pricing, Question and Method (2/2)

- Networks
  - an FFN for the weights $\omega$ and an adversary $g$ that builds the hardest test assets
  - a long short-term memory network (LSTM) that turns the macro history into 4 states
  - the full model is labelled GAN (Generative Adversarial Network)
- Chronological split: training 1967–1986, validation 1987–1991, test 1992–2016
- Benchmarks
  - the unregularised (LS) or elastic net (EN) linear SDF
  - the tangency portfolios of the Fama–French three- and five-factor models (FF3, FF5, Fama & French (2015))
- Configurations ranked by the validation Sharpe ratio
  - data and tuning in the Appendix

# Deep Learning in Asset Pricing: the Benchmarks after 2016 (1/2)



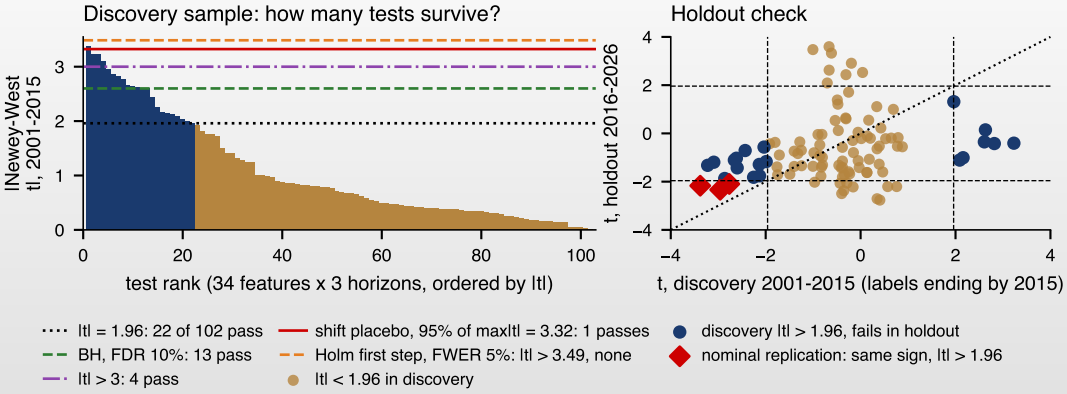

MFM_ch13_sdf_replication

# Deep Learning in Asset Pricing: the Benchmarks after 2016 (2/2)

- Same fixed weights, January 1992 – August 2026
  - each series scaled to the volatility of the market excess return on 1992–2016
  - monthly SR 2017–2026 ($T = 116$, SE $\approx 0.09$): FF5 0.06, FF3 0.03, market 0.24
  - Mkt-RF in the chart: the market excess return; on 1992–2016: 0.22, 0.20, 0.15
- Replicable with public data: the Fama–French benchmarks
  - not replicable: the stock-level GAN, LS and EN (the panel of the Center for Research in Security Prices, CRSP, is not in the course data)
  - the post-2016 fall is a warning, not yet a test, and the same question applies to a GAN frozen in 1991 (details in the Appendix)

MFM_ch13_sdf_replication

# Mini-Case: Thresholds and the Holdout (1/2)




MFM_ch13_ai_discovery

# Mini-Case: Thresholds and the Holdout (2/2)

- Left: the 102 discovery $|t|$ in decreasing order
- Right: discovery vs. holdout $t$
  - the three nominal replications (red) are all short-term reversal; none passes Holm at 5%
  - the significant VIX features collapse towards zero

MFM_ch13_ai_discovery

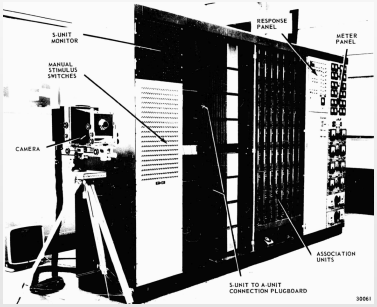
# Further Study on Quantinar

## Advanced courses on quantinar.com

- Random Forests
- Shapley Values
- Shapley Curves: A Smoothing Perspective
- Machine Learning in Financial Risk
- Introduction to Reinforcement Learning
- Generative Adversarial Networks
- MSCA Digital: The Need for Explainable AI
- Business Analytics and Data Science
- Model Selection Criteria
- PAC Learning

Back

# From the Perceptron to Deep Blue




Mark I Perceptron, operator's manual (1960)

**Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons**



IBM Deep Blue, Computer History Museum

**Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons**

- **Perceptron** Rosenblatt (1958)
  - a learning rule for a single artificial neuron
  - the Mark I: hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov (Campbell, Hoane & Hsu (2002))
- Deep Blue searched with hand-built rules
  - modern ML (this chapter) *learns* the rules from data

Back

# The In-Sample $R^2$ of a Predictor without Skill

- A monthly $R^2$ of 0.5% is economically large
  - but sampling noise is of the same order
- In-sample OLS $R^2$ of one predictor with no skill: $nR^2 \approx \chi^2_1$
  - intercept, independent errors, $n \approx 300$ months
  - $n$: number of months; $\chi^2_1$: chi-squared distribution with 1 degree of freedom, mean 1 and variance 2
  - so $R^2$ has mean $1/n \approx 0.3\%$ and standard deviation $\sqrt{2}/n \approx 0.5\%$
- $R^2_{OOS}$ can be negative and has no such null distribution
  - it needs its own test (Clark–West)

Back

## Worked Example: How a Forest Averages Its Trees

- Random forest of the case study (400 trees, minimum leaf size 100)
  - trained on the first four purged folds
- Prediction for the last observation (11 September 2026)
  - $P$(up in 5 days) from each tree
  - individual trees range from 0.31 to 0.88 (standard deviation 0.11)
  - 81.5% of the trees vote "up" ($p > 0.5$)
  - forest probability = average of the 400 trees = 0.608
- Single trees disagree strongly
  - the average is far more stable
- Stability is not accuracy
  - the forest still does not beat "always up" (case study)

Back

# Sampling Financial Data: Bars

## Time and tick bars

- A **bar** = one row summarising a sampling interval
  - open, high, low, close, volume
- Time bars: every fixed interval (1 minute, 1 day)
  - oversample quiet periods
  - fat-tailed returns
- Tick bars: one bar every $N$ transactions

## Volume, dollar and information bars

- One bar every $V$ shares or every $D$ dollars traded
  - sampling by **information arrival**
- Returns closer to i.i.d. draws from the Normal distribution
  - Clark (1973); Ané & Geman (2000)
- Imbalance / runs bars
  - sample when order flow becomes one-sided

- This chapter uses daily time bars
  - tick data (crypto exchanges) in Chapters 10 and 16

Back

# Fractional Weights and Their Asymptotics

## Definition 6 (Fractional difference operator)

- For a real order $d$:

$$(1-B)^d = \sum_{k=\mathbf{0}}^{\infty} \binom{d}{k} (-B)^k, \qquad \tilde{X}_t = \sum_{k=\mathbf{0}}^{\infty} w_k X_{t-k}$$

- Weights: $w_{\mathbf{0}} = 1$, $w_k = -w_{k-\mathbf{1}} \frac{d-k+\mathbf{1}}{k}$
  - for $d \notin \{0, 1, 2, \dots\}$: $w_k = \frac{\Gamma(k-d)}{\Gamma(-d)\,\Gamma(k+\mathbf{1})}$ and $w_k \sim \frac{k^{-(\mathbf{1}+d)}}{\Gamma(-d)}$ as $k \to \infty$

- $\binom{d}{k} = d(d-1)\cdots(d-k+1)/k!$: the binomial coefficient, defined for any real $d$
  - $\Gamma$: the gamma function, $\Gamma(k+1) = k!$
- The weights decay like $k^{-(\mathbf{1}+d)}$, more slowly for smaller $d$
- For $0 < d < 1$: $\sum_{k=\mathbf{0}}^{\infty} w_k = (1-1)^d = 0$
  - it follows from the binomial series at $B = 1$, which converges for $d > 0$
  - $d = 0$: $w_{\mathbf{0}} = 1$ and the rest zero, so the sum is 1; $d = 1$: $w = (1, -1)$, sum 0
  - for $d < 0$ the series of weights does not converge

Back

# Local Whittle and Exact Local Whittle (1/2)

- Periodogram: the sample estimate of the spectral density
  - at the Fourier frequency $\lambda_j = 2\pi j/n$

$$I(\lambda_j) = \frac{1}{2\pi n}\Big|\sum_{t=1}^{n} X_t\, e^{-\mathrm{i}\lambda_j t}\Big|^2$$

- Local Whittle (Robinson (1995)): choose $\delta$ to fit $f(\lambda) \approx G\lambda^{-2\delta}$
  - on the first $m$ frequencies

$$\hat{\delta} = \arg\min_{\delta}\ \log\Big(\frac{1}{m}\sum_{j=1}^{m}\lambda_j^{2\delta} I(\lambda_j)\Big) - \frac{2\delta}{m}\sum_{j=1}^{m}\log\lambda_j, \qquad \sqrt{m}\,(\hat{\delta}-\delta) \xrightarrow{d} N\big(0, \tfrac{1}{4}\big)$$

  - $n$: sample size; $m$: number of frequencies used; i: the imaginary unit; $G$ has been profiled out
  - $\xrightarrow{d}$: convergence in distribution; hence the standard error $1/(2\sqrt{m})$

Back

# Local Whittle and Exact Local Whittle (2/2)

- Exact local Whittle (ELW, Shimotsu & Phillips (2005); unknown mean: Shimotsu (2010))
  - periodogram of $(1-B)^{\delta} X_t$, valid also for non-stationary $\delta \geq \frac{1}{2}$
- Why the FFD series looks like $\hat{\delta} \approx 0.75$
  - above frequency $1/K$ the truncated filter acts as $(1-B)^{0.25}$
  - only $n/(2\pi K) \approx 2$ of the $m = 294$ frequencies lie below $1/K$, where the unit root returns
  - $K$: the FFD window length (444 lags)
  - the estimate is the apparent memory over the band, not the integration order of the series

MFM_ch13_ffd

Back

# Worked Example: Purging with 10 Observations

- Observations $t = 0, \ldots, 9$
  - label of $t$ ends at $t + 2$ (capped at 9)
- 5 contiguous folds of 2; embargo = 1 observation

| Test | Removed | Train |
|---|---|---|
| 0, 1 | 2, 3, 4 | 5–9 |
| 2, 3 | 0, 1, 4, 5, 6 | 7–9 |
| 4, 5 | 2, 3, 6, 7, 8 | 0, 1, 9 |
| 6, 7 | 4, 5, 8, 9 | 0–3 |
| 8, 9 | 6, 7 | 0–5 |

- Test $\{4, 5\}$
  - labels of 2 and 3 end at 4 and 5 $\Rightarrow$ purged
  - 6 and 7 start on or before day 7, when the last test label ends $\Rightarrow$ purged
  - the embargo removes 8, the first observation after day 7
- With overlapping labels and the embargo, an interior fold loses $2h + e = 5$ of 10 observations

Back

# When Is K-Fold Valid for Time Series?

- Bergmeir, Hyndman & Koo (2018): standard $K$-fold CV is valid for a **purely autoregressive** model
  - $y_t = g(y_{t-1}, \ldots, y_{t-p}) + \varepsilon_t$ with serially uncorrelated errors
  - $g$: any, possibly non-linear, function of the last $p$ values; $\varepsilon_t$: error
  - the errors in the test fold are uncorrelated with the training errors, so the CV error is unbiased for the OOS error
- It fails when the **errors** are serially correlated
  - overlapping labels: $h$-day forward returns share $h-1$ daily returns
  - persistent features that are not in the model's lag set (rolling means, levels such as the FFD log-price)
  - under-specified dynamics (too few lags)
- The leakage experiment isolates exactly this
  - 20-day labels and 20-day rolling features
- Check before choosing: the autocorrelation of the CV residuals
  - if it is significant, purge and embargo

Back

# Walk-Forward and Combinatorial Purged CV

## Walk-forward (WF)

- Mimics real-time deployment
  - no future data in training
- But a **single path**
  - high variance, easy to overfit by repeated trials

## Combinatorial purged CV (CPCV)

- $N$ contiguous groups
  - test on all $\binom{N}{k}$ combinations of $k$ groups, with purging and embargo
- $\varphi = \frac{k}{N}\binom{N}{k}$: the number of complete backtest paths
  - e.g. $N = 6$, $k = 2$: 15 splits, 5 paths
- A **distribution** of Sharpe ratios instead of one number

- Each group is tested $\varphi$ times, so the test predictions form $\varphi$ full histories

Back

# Testing Directional Accuracy: Pesaran–Timmermann

- $H_0$: the predicted sign $x_t$ is independent of the realised sign $y_t$ (Pesaran & Timmermann (1992))
  - $\hat{P}$, $\hat{P}^*$, $\hat{p}_y$, $\hat{p}_x$ as on the main slide, $n$: number of forecasts

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\widehat{V}(\hat{P}) - \widehat{V}(\hat{P}^*)}} \xrightarrow{d} N(0,1), \qquad \widehat{V}(\hat{P}) = \frac{\hat{P}^*(1-\hat{P}^*)}{n}$$

  - $\widehat{V}(\hat{P}^*) = \left[(2\hat{p}_y - 1)^2 \hat{p}_x(1-\hat{p}_x) + (2\hat{p}_x - 1)^2 \hat{p}_y(1-\hat{p}_y)\right]/n + 4\hat{p}_y\hat{p}_x(1-\hat{p}_y)(1-\hat{p}_x)/n^2$
  - $\widehat{V}(\cdot)$: estimated variance under $H_0$; $\hat{P}^*$ is itself estimated, hence the second term
- PT conditions on the marginal frequencies automatically
  - no need to guess the baseline
- Logit of the case study: $\hat{p}_y = 0.581$, $\hat{p}_x = 0.983$
  - one-sided $p = 0.79$ for $PT = -0.80$

MFM_ch13_inference

Back

# ROC AUC and Its Sampling Uncertainty

- Receiver operating characteristic (ROC) curve, over all thresholds
  - true positive rate $TP/(TP+FN)$ against false positive rate $FP/(FP+TN)$
- Area under the curve (AUC)
  - $\text{AUC} = P(\hat{p}_{\mathbf{pos}} > \hat{p}_{\mathbf{neg}}) + \frac{1}{2}P(\hat{p}_{\mathbf{pos}} = \hat{p}_{\mathbf{neg}}) = U/(n_{\mathbf{1}}n_{\mathbf{0}})$
  - $\hat{p}_{\mathbf{pos}}$, $\hat{p}_{\mathbf{neg}}$: predicted probabilities of a randomly drawn positive (up) and negative (down) observation
  - $n_{\mathbf{1}}$, $n_{\mathbf{0}}$: numbers of positives and negatives; $U$: Mann–Whitney statistic, the number of (positive, negative) pairs ranked correctly, ties counted $\frac{1}{2}$
- DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)): from the placement values
  - placement value: the share of the other class that each observation outranks
  - assumes independent observations
- Logit, pooled out-of-fold AUC $= 0.480$
  - DeLong 95% CI [0.466; 0.494]: "significantly worse than random"
  - block bootstrap 95% CI [0.456; 0.505]: noise around 0.5
- An AUC below 0.5 is anti-ranking only if it survives a test that respects the dependence

MFM_ch13_inference

Back

# Giacomini–White, Statistic and Rolling Window

- Giacomini & White (2006): the null concerns the forecasting **method**, not the population model
  - method: model + estimation window + refitting rule

$$H_0 : \mathbb{E}\big[d_{t+\tau} \mid \mathcal{F}_t\big] = 0 \;\Rightarrow\; \mathbb{E}\big[\mathrm{h}_t\, d_{t+\tau}\big] = 0, \qquad GW = n\,\bar{\mathrm{Z}}^{\top}\hat{\Omega}^{-1}\bar{\mathrm{Z}} \xrightarrow{d} \chi^2_q$$

- Notation
  - $d_{t+\tau}$: loss differential of the two forecasts made at $t$ for horizon $\tau = 5$ days; $\mathcal{F}_t$: information available at $t$
  - $\mathrm{h}_t$: test functions known at $t$ (here 1 and the lagged loss differential, $q = 2$); $\mathrm{Z}_t = \mathrm{h}_t d_{t+\tau}$
  - $\bar{\mathrm{Z}}$: mean of $\mathrm{Z}_t$ over the $n$ forecasts; $\hat{\Omega}$: HAC covariance matrix of $\mathrm{Z}_t$
- Reading: under $H_0$ no information at $t$ predicts which method will do better
- The theory requires a bounded (rolling) estimation window
  - so that estimation noise never vanishes
  - an expanding window lies outside the theorem
- S&P 500 logit, rolling 8-year window: GW $= 26.7$ ($p < 0.001$)
  - the logit forecasts worse than the constant

MFM_ch13_inference

Back

# Shapley Values: the Coalition Formula

## Definition 7 (Shapley value of feature $j$ (SHAP, Lundberg & Lee (2017)))

- Average marginal contribution of $j$ over all coalitions $S \subseteq \{1, \ldots, p\} \setminus \{j\}$:

$$\phi_j = \sum_S \frac{|S|!\,(p - |S| - 1)!}{p!} \big[v(S \cup \{j\}) - v(S)\big], \qquad \hat{f}(\mathsf{x}) = \phi_{\mathbf{0}} + \sum_j \phi_j$$

- Interventional SHAP: $v(S) = \mathbb{E}[\hat{f}(\mathsf{x}_S, \mathsf{X}_{\bar{S}})]$
  - averaging $\hat{f}$ with the features outside $S$ drawn from a background sample
  - $\phi_{\mathbf{0}} = v(\emptyset)$ = average prediction

- Notation
  - $p$: number of features; $S$: a coalition, a subset of the other features, $|S|$ its size, $\bar{S}$ the features outside it
  - $v(S)$: prediction when only the features in $S$ are known; $\hat{f}(\mathsf{x})$: model prediction for $\mathsf{x}$
- The weight of $S$ is a probability
  - the probability that exactly the features in $S$ come before $j$ in a random ordering of the $p$ features
- Conditional SHAP uses $v(S) = \mathbb{E}[\hat{f}(\mathsf{X}) \mid \mathsf{X}_S = \mathsf{x}_S]$
  - the two differ when features are dependent

Shapley Values

Back

## Reading Feature Importances

- Remedies for correlated features
  - permute a cluster's columns together, with the same row permutation, or refit without the feature
  - conditional and time-block permutations are separate variants
- Importances are estimates and need their own uncertainty analysis
  - honest trees give asymptotically Normal forest *predictions* and CIs (Wager & Athey (2018))
  - they do not give CIs for importances
- **Backtesting is not a research tool**, importance analysis is: López de Prado (2018)

Back

# Double Selection and DML, Penalty and Conditions (1/2)

- Partially linear model: $\mathbb{E}[\varepsilon_t \mid d_t, \mathrm{x}_t] = 0$, $\mathbb{E}[v_t \mid \mathrm{x}_t] = 0$, $\mathrm{Var}(v_t) > 0$
  - $g$, $m$ approximately sparse
  - the errors are unpredictable from the regressors; $d_t$ has variation of its own beyond the controls
- Post-double selection: the heteroskedasticity-robust standard error is valid (Belloni, Chernozhukov & Hansen (2014))
  - under independent sampling, regularity conditions and sparsity $s^2 \log^2(p \vee n)/n \to 0$
  - $s$: number of controls that matter; $p$: number of controls; $p \vee n = \max(p, n)$
- LASSO objective, with feature-specific penalty loadings

$$\frac{1}{n}\sum_t (y_t - \mathrm{x}_t^\top \beta)^2 + \frac{\lambda}{n}\sum_j \hat{\psi}_j |\beta_j|, \qquad \lambda = 2c\sqrt{n}\,\Phi^{-1}(1 - \gamma/(2p))$$

  - plug-in constants $c = 1.1$, $\gamma = 0.1/\log n$; loadings $\hat{\psi}_j^2 = \frac{1}{n}\sum_t x_{tj}^2 \hat{e}_t^2$, $\hat{e}_t$ the residuals of the previous step, iterated

Back

# Double Selection and DML, Penalty and Conditions (2/2)

- DML (Chernozhukov et al. (2018)): with $\ell(x) = \mathbb{E}[y \mid x]$, it suffices that
  - with consistency and regularity:

$$\|\hat{m} - m\| \left(\|\hat{\ell} - \ell\| + \|\hat{m} - m\|\right) = o_P(n^{-1/2})$$

  - $\|\hat{m} - m\|$: root mean squared estimation error of the learner for $m$; same for $\ell$
  - $o_P(n^{-1/2})$: tends to zero in probability faster than $1/\sqrt{n}$
  - e.g. both errors $o_P(n^{-1/4})$: each learner may converge more slowly than a parametric model, because only the product of the errors matters

Back

## Machine Learning for Inference in Asset Pricing

- Feng, Giglio & Xiu (2020): does a new factor add pricing information given hundreds of existing ones?
  - double selection of the controlling factors from the zoo, then a valid $t$-test of the new factor's risk price
  - the inferential answer to Harvey, Liu & Zhu (2016), who raise the hurdle to $t > 3$
- Heterogeneous effects: honest causal forests (Wager & Athey (2018))
  - they give CIs for effects that vary with covariates
- Three rules
  - prediction and inference are different targets: a good predictor can give a biased $\hat{\theta}$
  - orthogonalise before you test, and cross-fit with folds that respect time
  - report CIs; "selected" or "important" is not "significant"

Back

# Deep Learning for Sequences

## Recurrent networks (LSTM; GRU, gated recurrent unit)

- Hidden state carries memory: $h_t = g(h_{t-1}, x_t)$
  - $g$: a non-linear function with trained weights, $x_t$: input at $t$
- Gates control what is kept or forgotten
  - mitigating vanishing gradients
- Used for volatility, limit-order-book and intraday forecasting

## Transformers

- Self-attention: $\text{softmax}\left(QK^\top/\sqrt{d_k}\right)V$
  - query, key, value matrices $Q, K, V$ (linear maps of the inputs)
  - $d_k$: key dimension
  - softmax turns each row of scores into weights that sum to 1
- Long-range dependence, parallel training
- Backbone of large language models (Chapter 15)

Back

# Time-Series Foundation Models

## Pre-trained forecasters

- Pre-trained on large collections of series, used **zero-shot** (without re-training)
  - Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
- Useful baselines for volatility and volume forecasting
- For returns they inherit the same signal-to-noise limits
- Pre-training data may overlap your test period
  - a new form of leakage

- Covered in depth in Chapter 14
  - with the risk-forecast evaluation tools of Chapter 8

Back

# Reinforcement Learning, Generative Models, Explainable AI, Risk

## Reinforcement learning (RL)

- A policy $\pi(a \mid s)$ chooses actions $a$ in states $s$ to maximise cumulative reward
- Optimal execution, market making, deep hedging: Buehler et al. (2019)

## Explainable AI

- Required by regulators (EU AI Act, credit scoring, model risk)
- SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

## Generative models

- GANs, variational autoencoders (VAEs), diffusion models
  - synthetic scenarios for stress tests

## ML for risk

- Value-at-Risk (VaR): conditional quantiles (quantile forests, quantile-loss boosting)
- Expected Shortfall (ES): average of fitted tail quantiles or a joint VaR–ES loss
- Volatility: a squared-return target (Chapters 5 and 9)

- **Quantinar:** Reinforcement Learning · GANs · Explainable AI · ML in Financial Risk

Back

# Deep Learning in Asset Pricing, Data (Section III.A)

- Monthly returns of all CRSP stocks, January 1967 – December 2016
  - excess over the one-month Treasury-bill rate
- 46 firm characteristics in six groups (Table A.II)
  - past returns, investment, profitability, intangibles, value, trading frictions
  - Only stock-months with all 46 available: about 10,000 of 31,000 stocks
  - ranked each month into cross-sectional quantiles
- 178 macroeconomic series, transformed to stationarity
  - 124 of FRED-MD, the St. Louis Fed's monthly macroeconomic database (McCracken & Ng (2016))
  - the 46 monthly medians of the characteristics, 8 predictors of Welch & Goyal (2008)

Back

# Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (1/2)

- Objective (eq. 3) and networks: main slide
  - the adversary $g$ uses 8 instruments
  - three steps: unconditional SDF, maximise over $g$, minimise again (Section II.D)
- Tuning: 384 configurations ranked by the validation Sharpe ratio (SR)
  - chosen: 2 layers of 64 units, dropout, Adam; ensemble of 9 fits (Table A.VIII)
  - loadings: a second FFN fits $E_t[R^e_{t+\mathbf{1},i} F_{t+\mathbf{1}}] \propto \beta_{t,i}$
- The linear LS/EN benchmark: SDF $\omega_{t,i} = \theta^\top I_{t,i}$
  - on long and short characteristic legs
  - $\omega_{t,i}$: weight of stock $i$; $I_{t,i}$: its characteristics; $\theta$: one coefficient per characteristic, the same for all stocks and months

Back

# Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (2/2)

- Metrics (II.F): the SR of $F$ and two fit measures
  - $\hat{\epsilon}_{t+1,i}$: pricing error, the part of $R^e_{t+1,i}$ not explained by $\hat{\beta}_{t,i} F_{t+1}$
- Explained variation (EV)
  - one minus the average squared pricing error over the average squared excess return

$$\text{EV} = 1 - \frac{\overline{\hat{\epsilon}^2}}{\overline{(R^e)^2}}$$

  - overline: time average of the monthly cross-sectional means; EV $= 1$: the factor explains all variation
- Cross-sectional $R^2$: how well the model explains average returns

$$\text{XS-}R^2 = 1 - \frac{\frac{1}{N}\sum_i \frac{T_i}{T}\left(\bar{\hat{\epsilon}}_i\right)^2}{\frac{1}{N}\sum_i \frac{T_i}{T}\left(\bar{R}^e_i\right)^2}$$

  - $\bar{\hat{\epsilon}}_i$, $\bar{R}^e_i$: time averages for stock $i$; $T_i$: months of data on stock $i$, $T$: months in the sample

Back

# Deep Learning in Asset Pricing, Results (1/2)

- Source: Chen, Pelger & Zhu (2024), *Management Science*
- Out-of-sample SR of the SDF (Table I), annualised: GAN 2.6, EN 1.7, FFN 1.5
  - monthly values on the main slide
- Individual stocks (Table I)
  - EV 8% (GAN) vs. 3–4%; cross-sectional $R^2$ 23% vs. 14–19%
  - Reading: the no-arbitrage objective improves both fits, although it maximises neither
- Deciles sorted on the loadings $\beta$ (Table II, test period, annualised)
  - top minus bottom 0.39; FF5 alpha 0.39, $t = 9.96$
  - the GRS test rejects the CAPM (capital asset pricing model), FF3 and FF5 ($p = 0.00$)
  - GRS (Gibbons–Ross–Shanken) test: $H_0$ all alphas of the decile portfolios are zero
  - Reading: loadings predict returns, linearly ($R^2 = 0.97$ of mean return on $\beta$, Figure 8), and the factor models do not price them

Back

# Deep Learning in Asset Pricing, Results (2/2)

- 46 sets of characteristic-sorted deciles (Table III)
  - GAN cross-sectional $R^2$ above 90% for all
  - EV short-term reversal 0.70 vs. 0.43 (EN), momentum 0.54 vs. 0.26
  - Reading: gains come from extreme deciles and interactions
- Macro states (Figure 6): the 178 raw series make the out-of-sample SR collapse
  - GAN loses about 10% without macro states and 20% with unconditional moments only
  - Reading: the dynamics of macro data must be modelled (LSTM), not fed as the last increment
- Largest stocks (Table IV, monthly SR)
  - models fitted on all stocks, evaluated on the subset
  - about 1,500 stocks GAN 0.41, FFN 0.30, EN 0.15
  - about 550 stocks 0.26, 0.24, 0.23
  - re-estimated on the 1,500: 0.42, 0.28, 0.42
  - Reading: the edge narrows sharply among the largest stocks; EN catches up when re-estimated on them

Back

# Deep Learning in Asset Pricing, Replication and Extension

- With the Kenneth French Data Library and FRED-MD
  - on the published split
  - Replicable: FF3 and FF5 tangency weights estimated on 1967–1986 (footnote 23), monthly SR on 1992–2016 next to Table A.VI
  - Not replicable with the course data: the published LS and EN (Tables I and III) use rank-weighted legs of 46 characteristics and stock-level betas
  - Analogue on portfolio data, an extension rather than a replication
  - the linear SDF ($\hat{\theta}_{LS}$ in closed form, $\hat{\theta}_{EN}$ with the objective of Section II.F) on halves (deciles 1–5, 6–10) of the 15 characteristics that the library sorts into deciles
  - its levels are not comparable with Tables I and III

Back

# Deep Learning in Asset Pricing, Limits and an Open Extension

- Limitations stated by the authors
  - The positivity of the SDF is not imposed (footnote 9)
  - only stock-months with all characteristics, which mainly drops very small stocks (Section III.A)
  - Frictions (Section III.I, Figure 17): annual SR 1.73 without the smallest 40% of stocks
  - 2.07 without the widest 40% bid–ask spreads
  - 1.87 without the lowest 40% turnover; see also Avramov, Cheng & Metzker (2023)
  - In-sample metrics exceed the out-of-sample ones, a sign of some overfitting (Section III.C)
  - the hidden states are not comparable across fits, so they cannot be averaged (Section II.E)
- Open extension: freeze the 1967–1991 design
  - evaluate 2017–2026, after the paper's sample
  - does the linear no-arbitrage SDF still beat FF5?
  - do LSTM states of FRED-MD add to it?

Back

# Deep Learning in Asset Pricing, Reading the Two Charts

- GAN 0.75 vs. FF5 0.22
  - with Lo SEs 0.065 and 0.058, the SE of the difference is at most 0.12 whatever their correlation
  - the gap exceeds 4 SEs (i.i.d. Normal returns)
  - a paired test on aligned returns (Ledoit & Wolf (2008)) is sharper
- The fixed FF5 weights load mainly on profitability and investment
  - RMW 0.42, CMA 0.30 (weights scaled to $\sum |w| = 1$)
  - FF3 puts 0.61 on HML
  - both combinations were chosen on 1967–1986

MFM_ch13_sdf_replication

Back

# The Discovery Loop, Step by Step (1/2)

- **Literature**
  - `"Summarise the evidence on how much published return predictability survives out of sample and after publication: DOI or arXiv id, sample, headline decay estimate."`
  - Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search
- **Hypotheses**
  - `"For a feature that predicts S&P 500 returns in 2001-2015, give competing explanations (risk premium, reversal, microstructure, data mining) and what each predicts for 2016-2026 and for other indices."`
- **Data and code**
  - `"Write Python that builds 34 technical features from the course's daily S&P 500 and VIX closes with information up to day t only, and reports Newey-West t-statistics for 1-, 5- and 20-day returns in 2001-2015."`
  - Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

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# The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance
  - `"Write the pre-analysis plan: the full list of features and horizons, the holdout, the corrections (Holm, BH, max-t) and the result that would count as a failure."`
  - Placebo: shifting the target circularly (250–3,000 days) breaks its alignment with the features but keeps each series' own dependence
  - a stationary target does not give the joint null of all zero slopes: the shift also changes the features' link with conditional volatility
  - hence a heuristic yardstick; an alternative: a joint bootstrap of the scores, centred under the null, with explicit assumptions
- **Critique**
  - `"Act as a referee: which of these significant features are one effect counted several times, and which are known anomalies rather than new ones?"`
- **Report**: all $M$ tests, all trials including the discarded ones
  - DSR and PBO for any strategy built on them

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# Acronyms used in this material (1/3)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AUC**: Area Under the (ROC) Curve
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **C-ENet**: Combination Elastic Net
- **CAPM**: Capital Asset Pricing Model
- **CBOE**: Chicago Board Options Exchange
- **CF**: Cash Flow (in CF/P: cash flow to price)
- **CI**: Confidence Interval
- **CMA**: Conservative Minus Aggressive (investment factor)
- **CPCV**: Combinatorial Purged Cross-Validation
- **CRSP**: Center for Research in Security Prices
- **CSCV**: Combinatorially Symmetric Cross-Validation
- **CV**: Cross-Validation
- **CW**: Clark–West (test for nested models)
- **DM**: Diebold–Mariano (test)
- **DML**: Double/Debiased Machine Learning
- **DSR**: Deflated Sharpe Ratio
- **ELW**: Exact Local Whittle (estimator)
- **ENet**: Elastic Net
- **ES**: Expected Shortfall
- **EU**: European Union
- **EV**: Explained Variation
- **EWMA**: Exponentially Weighted Moving Average
- **FDR**: False Discovery Rate
- **FF3**: Fama–French three-factor model
- **FF5**: Fama–French five-factor model
- **FFD**: Fixed-width window Fractional Differentiation
- **FFN**: FeedForward Network
- **FN**: False Negative

## Acronyms used in this material (2/3)

- **FP**: False Positive
- **FRED-MD**: Federal Reserve Economic Data, Monthly Database (McCracken–Ng)
- **FWER**: Family-Wise Error Rate
- **GAN**: Generative Adversarial Network
- **GB**: Gradient Boosting (the seminar's name for GBM)
- **GBM**: Gradient Boosting Machine (here not geometric Brownian motion)
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **GRU**: Gated Recurrent Unit
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HARKing**: Hypothesising After the Results are Known
- **HML**: High Minus Low (value factor)
- **IBM**: International Business Machines (Corporation)
- **IS**: In-Sample
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **LIME**: Local Interpretable Model-agnostic Explanations
- **LLM**: Large Language Model
- **LS**: Linear SDF estimated by unregularised Least Squares
- **LSTM**: Long Short-Term Memory (network)
- **LW**: Local Whittle (estimator)
- **MA**: Moving Average
- **MA20**: 20-day Moving Average
- **MDA**: Mean Decrease Accuracy
- **MDI**: Mean Decrease Impurity
- **ML**: Machine Learning
- **MLP**: MultiLayer Perceptron
- **MSCA**: Marie Skłodowska-Curie Actions

## Acronyms used in this material (3/3)

- **OLS**: Ordinary Least Squares
- **OOS**: Out-Of-Sample
- **PAC**: Probably Approximately Correct (learning)
- **PBO**: Probability of Backtest Overfitting
- **PCR**: Principal Component Regression
- **PLS**: Partial Least Squares
- **PSR**: Probabilistic Sharpe Ratio
- **PT**: Pesaran–Timmermann (test of directional accuracy)
- **RC**: Reality Check (White's test for data snooping)
- **ReLU**: Rectified Linear Unit, $\max(0, z)$
- **RF**: Random Forest
- **RL**: Reinforcement Learning
- **RMW**: Robust Minus Weak (profitability factor)
- **ROC**: Receiver Operating Characteristic (curve)
- **RSI**: Relative Strength Index
- **SD**: Standard Deviation
- **SDF**: Stochastic Discount Factor
- **SE**: Standard Error
- **SHAP**: SHapley Additive exPlanations
- **SMB**: Small Minus Big (size factor)
- **SPA**: Superior Predictive Ability (test)
- **SR**: Sharpe Ratio
- **TN**: True Negative
- **TP**: True Positive
- **US**: United States
- **VAE**: Variational AutoEncoder
- **VaR**: Value-at-Risk
- **VIX**: CBOE Volatility IndeX
- **WF**: Walk-Forward (validation)

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